Wind Turbine Entry and Transmission Expansion: Evaluating Spatial Subsidies in Germany

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Abstract

Ambitious renewable energy targets create a spatial mismatch between electricity supply and demand by encouraging energy generation in resource-rich areas far away from demand centers. I study how spatial subsidy design and delayed transmission investment can alleviate this mismatch and how they interact. I develop and estimate a spatial model of onshore wind-turbine entry and combine it with a network-constrained dispatch model of the German electricity system with endogenous congestion management. I compare alternative spatial renewable subsidy designs, before and after a planned grid expansion. Spatially flat subsidies raise total modeled costs by about two percent on the existing grid but perform similarly to the current wind-productivity-adjusted subsidy after expansion, showing that locational correction is important during transition to a new grid. Congestion-adjusted support significantly improves the integration of wind generation, lowering costs by about two percent. Without grid expansion, it achieves 75 percent of the cost reduction obtained by combining the status quo subsidy with the planned grid expansion. I demonstrate that spatial subsidy design and transmission expansion are joint policy margins: congestion-adjusted support can partly substitute for grid investment, lowering costs and improving renewable integration during the transition while reducing the need for costly grid expansion.

1 Introduction

The green energy transition shifts electricity generation toward resource-rich areas that are often imperfectly interconnected with major demand centers. This creates a spatial mismatch between supply and demand that leaves the policy maker with three main options: expand transmission capacity to the resource-rich areas, incentivize generation in better-connected locations or combine both. This policy challenge arises worldwide and is particularly apparent in the European Union (EU). Grid operators plan to invest roughly EUR 72 billion per year through 2030 in transmission and distribution infrastructure (European Court of Auditors, 2025). Transmission expansion takes time, however, and the rapid growth of geographically concentrated and intermittent renewable generation is already straining existing networks, highlighting imperfect interconnection. Congestion management cost the EU EUR 4.2 billion in 2023, of which Germany accounted for approximately 60 percent (European Union Agency for the Cooperation of Energy Regulators, 2024). This shows that when renewable generation expands faster than the grid can adjust, electricity that is cheap to produce may nevertheless be costly to integrate. This paper evaluates how spatial subsidy design affects the geography of wind turbine entry, how it endogenously changes the cost of congestion management until the grid can expand and how it alters incentives to invest into the grid. I therefore answer the question how two policy margins—the spatial design of renewable support and planned transmission expansion—interact in reducing the spatial mismatch between energy supply and demand while accommodating a large wind build-out.

I study this interaction empirically by combining a spatial entry model of German onshore wind investment with a network-constrained electricity dispatch model allowing for endogenous congestion management. The two components are linked by translating the spatial allocation of wind turbines into electricity generation using turbine technology and local wind conditions. The resulting framework allows me to quantify how changes in spatial subsidy design affect the geography of wind investment, how the resulting generation is accommodated by the transmission network, and how these changes affect generation and congestion-management costs. Explicitly modeling the physical transmission network further allows me to evaluate how the same allocation of wind turbines performs before and after planned grid expansion.

For the entry model, I use project-level data on all onshore wind turbines entering between 2004 and 2023, including their exact location, turbine technology, installed capacity, and commissioning year. Building on the renewable entry model of Gonzales et al. (2023), I allow for heterogeneous turbine technology and persistent differences in entry costs across regions and cohorts. I aggregate wind projects to the county–cohort level and estimate an inverse entry equation relating expected discounted lifetime revenue to installed wind capacity, controlling for region and entry-cohort fixed effects.

For the dispatch model, I divide Germany into four zones corresponding to the regions served by its four transmission system operators (TSOs). I represent the observed transmission lines connecting these zones and their physical properties, allowing for transmission constraints between

but not within TSO regions. Building on the network-constrained dispatch framework of Bohn et al. (1984) and Arkolakis and Walsh (2023), I incorporate endogenous congestion management following the institutional sequence of the German wholesale electricity market. I simulate the model using data on electricity demand, renewable and conventional generation, imports, generation availability, and marginal costs. In the first stage, the national market minimizes generation costs without accounting for internal transmission constraints. In the second stage, system operators adjust the initial generation schedule to accommodate these constraints. This involves redispatch, which shifts conventional electricity generation between locations, and curtailment, which reduces wind generation when it cannot be accommodated by the network. Both arise from transmission congestion and capture the additional costs of integrating electricity generation into the grid. I validate the dispatch model against observed redispatch and curtailment data for Germany in 2023.

I use this empirical framework to evaluate three spatial renewable subsidy designs before and after a planned grid expansion. Under the status quo, subsidies are adjusted according to local wind productivity, with lower productivity leading to higher support. I compare this policy with two alternative subsidy rules under the same cohort-specific government budget: a geographically flat subsidy and a subsidy adjusted by lagged locational marginal prices, i.e. a congestion-adjusted subsidy. The former mimics free entry under a common guaranteed support level while the latter incentivizes wind turbine entrants to take the system value of their location choice into account. The estimated entry model first determines where wind investment locates under each subsidy rule, after which the dispatch model maps this into generation and congestion costs. I then add the major north–south high-voltage direct current (HVDC) transmission lines — planned in the early 2010s and projected to be operational starting 2026 — to the observed network and operate each wind allocation on the expanded grid. The model therefore quantifies both how subsidy-induced spatial entry affects congestion before the grid adjusts and how it changes the benefits of subsequent transmission investment.

The paper has four main results. First, the counterfactuals show how subsidy design changes the geography of wind entry. Under the status quo, most of the onshore wind energy capacity is installed in the North of Germany close to the coast. Moving to a spatially flat support exacerbates this tendency shifting investment even more toward the most productive wind locations in the northern and coastal regions. It further lowers entries in the low wind speed southern regions of Germany. Congestion-adjusted support shifts entry away from locations with low marginal system value and toward higher-value parts of the grid. It increases northern investment where wind productivity remains sufficiently high while also inducing entry in the Southwest of the country where the system-value is large.

Second, on the existing grid, flat support raises electricity-system cost by about 1.9 percent relative to the status quo even though it lowers unconstrained generation cost and increases gross wind production. The gain from more productive sites is more than offset by substantially higher redispatch and curtailment. Congestion-adjusted support moves in the opposite direction. On

the existing grid it lowers total modeled electricity-system cost by about 1.8 percent and reduces redispatch and curtailment. Because Germany has no locational wholesale prices, the flat policy is a useful benchmark for free entry under a common guaranteed support level: entry responds to wind productivity and support but not to the local value of electricity. The result therefore shows that a spatial entry signal is important for keeping system costs low during the transition, when wind investment can respond well before transmission capacity is expanded. This is because the grid is already significantly congested. I view the estimated gain from congestion adjustment as conservative because the subsidy signal varies only across the four TSO regions, whereas the status quo support differentiates at a much finer spatial level.

Third, once the planned grid expansion is added, the cost difference between flat and status-quo support largely disappears. At the same time, the value of the same transmission investment depends strongly on the wind geography created by subsidy policy. The planned expansion is approximately cost-covering under the observed allocation but becomes substantially more valuable under flat support, where additional transfer capacity relieves the larger spatial mismatch created by entry into highly productive but transmission-constrained regions. Flat support and grid expansion are therefore complements: removing the spatial subsidy signal increases the amount of network capacity that is valuable later. Without any grid expansion, congestion-adjusted support achieves roughly 75 percent of the cost reduction obtained by combining the status quo subsidy with the planned grid investment. Because part of the spatial mismatch is already removed before transmission is expanded, the grid expansion is no longer cost-covering under congestion-adjusted support. The two instruments are therefore imperfect substitutes. This interaction also illustrates why evaluating transmission investment independently of renewable support can be misleading.

Fourth, congestion-adjusted support lowers the dispersion in locational marginal prices even on the existing grid, indicating improved interconnection through subsidy design alone. With grid expansion, locational prices converge further as additional transmission capacity relieves congestion.

From a policy perspective, the results show that subsidy design and transmission expansion should be determined jointly; improving locational incentives before new turbines are built can reduce the amount of costly network reinforcement required later while significantly lowering electricity system costs in transition. My paper has further direct implications for the plans of the German regulator to increase wind capacity by another 58 percent until 2030. The finding that the currently planned grid investment is approximately cost-covering under the observed subsidy regime implies that it is significantly underinvested for the planned wind expansion of additional 58 percent. As additional grid expansion takes more time than available, implementing congestion-adjusted subsidies in the meantime would help lowering system cost integrating more of the new wind generation until necessary grid adjustments can be made.

Germany provides a particularly suitable setting for studying the research question of this paper. The country combines a long history of large-scale wind deployment with a single national wholesale bidding zone, implying a common wholesale electricity price across the country, and a transmission

network inherited from an electricity system dominated by coal and nuclear generation. Wind resources are most abundant in the North, while major demand centers lie farther south and west. Planning for the north–south transmission expansion began in the early 2010s, reflecting the policy relevance of adapting this legacy network to a generation mix increasingly concentrated in wind-rich regions. Renewable policy has also responded directly to the geography of wind investment. German regulators have long differentiated wind subsidies according to local wind productivity, awarding higher subsidy rates to turbines at less productive locations. They have also temporarily restricted auction awards in export-constrained northern regions of Germany. These regulations illustrate that the spatial mismatch created by renewable entry has been a concern for policymakers.

The setting is further favorable for estimating the wind-entry response to spatial incentives. I observe twenty entry cohorts from 2004 to 2023, covering a long period of large-scale deployment. The Renewable Energy Sources Act (EEG) provides a twenty-year support horizon, allowing me to construct the expected lifetime revenues of each project rather than relying on contemporaneous revenues. Lifetime revenues vary substantially across locations and cohorts because wind resources differ, turbine technologies evolve, and support varies by location. This provides rich variation in lifetime revenue allowing me to control for persistent regional differences and common cohort shocks.

An additional institutional feature facilitates the identification of entry responses to lifetime revenues. Wind developers are largely insulated from the local congestion costs created by their entry. Germany has one national wholesale price, renewable generators have historically been largely compensated when curtailed, and developers do not bear the cost of reinforcing the existing transmission network. These regulations make the interaction between renewable siting and network congestion effectively one-sided. In particular, local congestion does not directly enter the lifetime revenue measure used in the entry model. This reduces an important potential confounding channel: high project revenue is not mechanically a proxy for favorable local transmission access.

The paper contributes to three strands of literature. First, it relates to renewable siting and locational investment incentives. The value of renewable generation varies with resource quality, displaced generation, and network access (Lin, 2016; Sexton et al., 2021). Lamp and Samano (2023) documents spatial misallocation of solar generation in Germany, including the role of transmission constraints. Schmidt and Zinke (2023) and Katzen and Leslie (2024) study distorted siting incentives under uniform or zonal electricity prices; related work examines differentiated support and network charges (Grimm et al., 2019; Tangerås and Wolak, 2019; Wagner, 2019). Other studies address policy uncertainty, technological progress, reliability, and intermittent generation (Ambec and Crampes, 2019; Chen, 2025; Covert and Sweeney, 2026; Elliott, 2026). I contribute an estimated response of German wind entry to spatial subsidies, allowing me to evaluate congestion-adjusted support within a uniform-price wholesale market rather than imposing a planner's allocation.

Second, the paper relates to transmission investment and renewable integration. Transmission affects renewable utilization and market efficiency (De Cannière, 2026; Dorsey-Palmateer, 2020; Fell et al., 2021; Hausman, 2025; LaRiviere and Lyu, 2022). Gonzales et al. (2023) estimates how

market integration induces renewable entry, while Doshi (2026) studies wind investment following grid expansion. Arkolakis and Walsh (2023) embeds electricity transmission in a spatial model of clean growth, and Johnston et al. (2026) examines interconnection barriers. I study the reverse policy channel: how subsidy-induced wind locations change congestion costs and the value of a planned grid expansion. Combining estimated entry with endogenous redispatch and curtailment, I quantify when and to which extent locational support complements or substitutes for transmission investment during grid adjustment.

Third, the paper connects to spatial and network economics, where the location of activity and the value of infrastructure depend on network frictions (Allen and Arkolakis, 2014). Fajgelbaum and Schaal (2020) studies investment across congested transportation links. In shipping, Brancaccio et al. (2017) shows that trade geography and vessel availability determine transport costs, while Brancaccio et al. (2024) shows how port investment creates congestion spillovers across the network. Unlike routable shipments, electricity follows physical network constraints. My contribution is to show how changing investment incentives at network nodes alters congestion and the returns to expanding links. The central contribution is therefore to connect private siting incentives to the physical transmission network and to show that the spatial design of renewable support changes not only current congestion costs but also the economic value of subsequent grid expansion.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 presents the institutional setting and constructs expected lifetime wind revenues. Section 4 estimates the onshore wind-entry model. Section 5 introduces the electricity-dispatch and pricing model. Section 6 evaluates the subsidy and transmission counterfactuals, and Section 7 concludes.

2 Data

I collect data on power generation plants in Germany, power curves of wind turbine models, wind speed, subsidies for wind turbines, electricity market outcomes, the physical properties of the electric grid and planned extension projects.

Power generation plants. For the whole of Germany I observe the universe of past and present power generation plants with their exact location. The dataset is publicly available and provided by the German regulator for electricity, gas, telecommunications, railway and post markets (Bundesnetzagentur hereafter BNA). The data set includes the beginning and end date of operation, the installed capacity and energy source. Crucially, for wind turbines I observe the precise model of wind turbine installed including its height and rotor diameter. Additionally, for each wind turbine the data set features an indicator that allows me to know in which auction they were granted their subsidy. For a subset of all onshore wind turbines it further includes the ratio between expected generated electricity at the actual and the same turbine model installed at a reference location defined by the regulator. This is called the Guetefaktor or hereafter quality factor. Its denominator is the reference yield which I also observe separately for the most common wind turbine models

provided by FGW (Association for Wind Energy and other Decentralised Energies). These measures are important to calculate the effective subsidy obtained. For offshore wind turbines I complement this with information on the coastline of Germany from the Bundesamt fuer Seeschifffahrt und Hydrographie (BSH) and data on water depth for North and Baltic Sea comes from the European Marine Observation and Data Network (EMODnet). Again, this is needed to calculate effective subsidies received. I complement the information on power generation plants with plant-specific efficiencies, emission factors, fuel and transportation costs which are used to construct marginal costs for 2023. The information is provided through the Merit Order Tool of the Institute of Energy Economics at the University of Cologne (EWI).

Power curves. For most models of the largest wind turbine manufacturers I observe the power curve. I obtain this data from The Wind Power which is a commercial data provider to the industry commonly used by planners when deciding on location and wind turbines for their wind project. The power curve is a function that maps given wind speed in meters per second into electricity produced. With location specific wind speed data it hence allows to project the expected electricity output at a given location for a specific wind turbine model. The power curves are important indicators of efficiency of a wind turbine and have significantly changed over the years with innovation by wind turbine manufacturers.

Wind speed. I work with average as well as hourly wind speed data. The former is in meters per second at the height of 100 meters in 250 x 250 meter rasters. The dataset is freely available, provided by the Global Wind Atlas and based on the ERA5 dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF) for the years 2008 to 2017. The hourly wind data comes from the same source but covers the year 2023 for a 0.25 by 0.25 degree resolution.

Wind turbine subsidies. For the years 2004 up until 2023 I have collected all feed-in tariffs for renewable energy plants as well as the quantity-averaged outcomes of the pay-as-bid auctions that have gradually superseded the feed-in tariffs starting 2017. This information comes from the transmission grid operators (TSOs).

Electrical grid. Information on all substations in Germany comes from JAO, a service provider for TSOs in Europe. The same is the case for transmission lines (high voltage) connecting these substations. The dataset also includes important properties of the lines such as their length, resistance and reactance. The grid information is extracted for the year 2024. I further collected the grid access points of all offshore wind parks in Germany. Information on projected costs, connecting substations and capacities of the planned high voltage direct current (hvdc) transmission cables has also been collected by hand directly from official documents of the TSOs.

Electricity market outcomes. The TSOs report the monthly average spot market prices for wind energy which takes into account that wind output is highly correlated and a zero marginal cost technology leading to generally lower prices obtained by wind turbine operators. This measure is needed to calculate the effective subsidy for the years in which a fixed feed-in tariff was not available anymore. I further use hourly data from BNA on realized renewable generation, imports/

exports and system load as well as load from hydro-pumped storage. From ENTSO-E I obtain actual hourly generation along with unavailabilities for all generation units with an installed capacity above 100MW. For the year 2023 I observe curtailment at the TSO-level accessible through the Netztransparenz initiative of the TSOs.

2.1 Wind speed distribution

Germany's geography is such that the North is more favorable for wind turbine deployment. At a higher level Germany is located in a west wind zone. The North of Germany is rather flat. In the North west of Germany there is the North Sea which gives rise to high wind speeds in this region. This is true to a slightly smaller extent in the North-East of Germany which is protected from the West wind by more (although flat) landmass than in the North-West. The Middle and South of the country are characterized by low mountain ranges leading to on average lower wind speeds but also to some regional areas of larger wind speed. These areas are generally less accessible, though. I show the average wind speed in Figure 1.

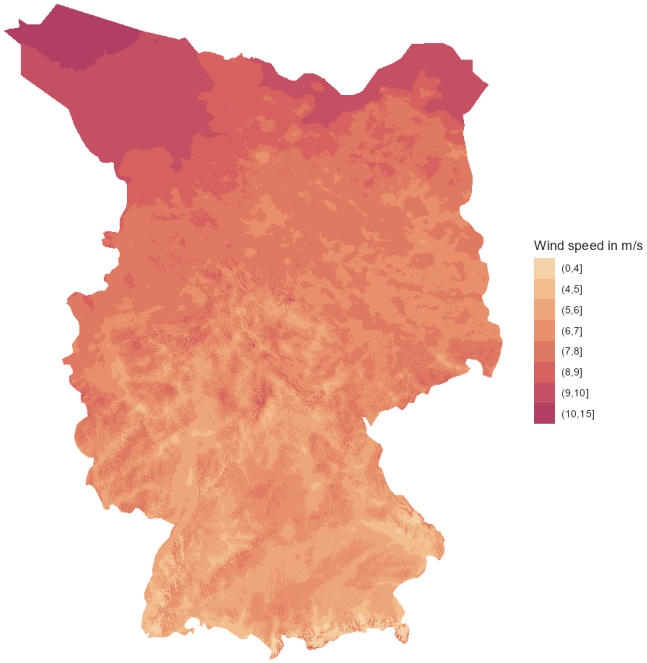
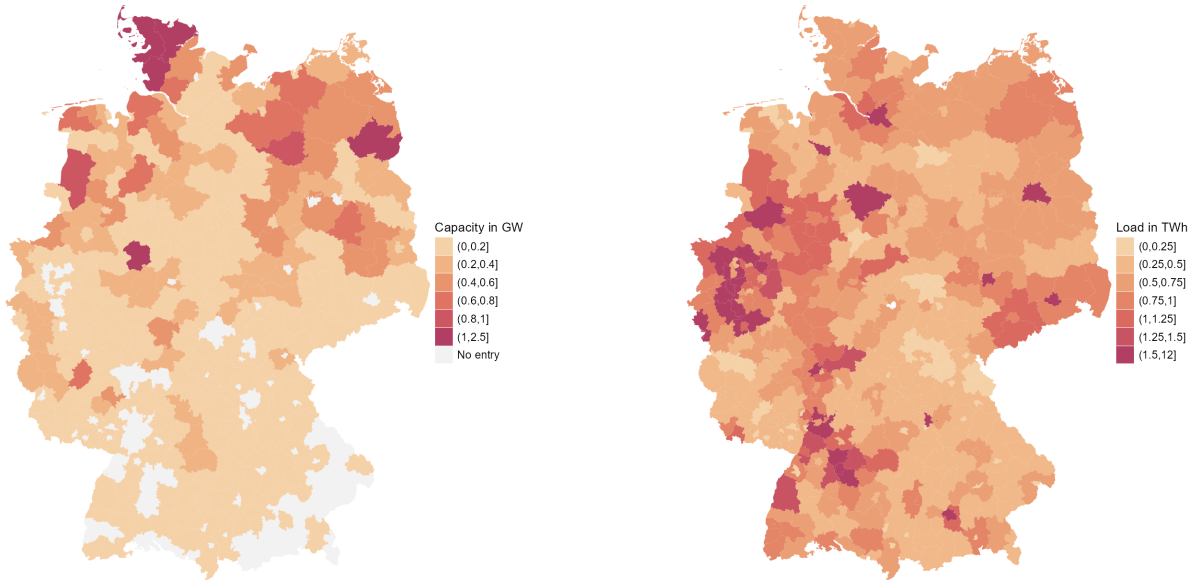


Figure 1. Average wind speed on- and offshore at a height of 100 meters

2.2 Wind turbine locations and load centers

In the following I focus on onshore wind turbine entry by county excluding offshore entry. In Figure 2 we can see that wind capacity installed is mainly concentrated in the very North. Especially some counties at the border to Denmark as well as the Netherlands stand out. We can further

observe that in the South there are multiple counties entirely unexplored which is also true for the West which is driven by high population density, inaccessible mountain ranges and regional political resistance. The installed wind capacity correlates strongly with regions of high average wind speeds. The figure further reflects policy decisions which will be explained in more detail later. The right panel shows that electricity demand is concentrated especially in the West and South West, geographically far from the predominant locations of installed wind capacity in the North.



(a) Cumulative installed onshore wind capacity

(b) Average yearly electricity load

Figure 2. Onshore wind capacity and electricity load

2.3 Grid congestion

The German electrical gird is historically constructed to connect the coal rich regions of the West and East, as well as nuclear power plants to the large demand centers in the West and South West. It has therefore little capacity to transport electricity from the very North to the rest of Germany. Looking at the cumulative wind generation together with the German coal and nuclear exit exactly this is now needed, though.

Table 1. Evolution of redispatch, curtailment and wind capacity

	Redispatch		**Curtailment**		**Total wind capacity**
	TWh	Mio. Euros	TWh	Mio. Euros	GW
2013	4.4	113	0.5	44	34.2
2017	18.5	392	5.5	610	56.1
2021	20.4	590	5.8	807	63.7

Table 1 shows the development of grid congestion together with wind turbine deployment. The German electricity market works in two stages where in a day-ahead market load and renewable generation are projected and the market clears without taking line constraints into account. On the actual day in any given moment demand and supply of electricity have to exactly equal in Germany (including imports and exports). So the electricity market is cleared again where the regulator minimizes the deviation from the day-ahead market clearing taking into account the potentially restricted grid. There are two common tools the regulator uses to achieve that. First, they can ramp down certain conventional power plants previously dispatched in the day-ahead clearing and ramp up others instead. When this is not possible anymore the operator can curtail renewable power plants. This means that they generate more than is actually injected into the system. This effectively creates wasted renewable energy. By regulation, though, renewable energy has priority in dispatch which is why there is redispatch before there is curtailment. The table shows that both measures scale with the installed wind energy capacity. This is extremely clear when comparing 2013 and 2017. Redispatch increased fourfold while curtailment rose by a factor of 10. At the same time cumulative wind capacity increased by a factor of 1.6. From 2017 to 2020 the regulator restricted wind turbine entry in the very far North which lead to disproportionately lower congestion measures comparing 2017 and 2021 even though wind capacity still increased significantly. Redispatch and curtailment increased by 10 and 5 percent, respectively, while cumululative wind capacity changed by 13 percent. This suggests that there is a spatial component to how wind turbine entry affects congestion which is the main focus of this paper. The monetary values further suggest that tackling the congestion issue can lead to significant relief in cost.

As a response to the poorly suited gird infrastructure facing increased wind turbine entry in the far North the regulator has planned three major transmission lines split into five projects which are planned to be operational from 2026 and 2028 onward. They are shown in Figure 3. We can see that two of them aim to alleviate the described issue connecting the North to demand centers in the West and South West.

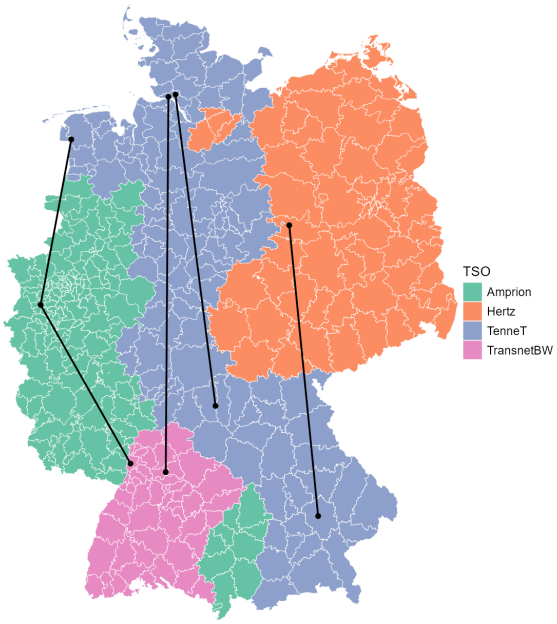


Figure 3. The five planned HVDC transmission lines and the TSOs

3 Institutional Context

Germany has pursued unusually ambitious wind-energy expansion for much of the past two decades, although the strength of this ambition has varied over time. Rapid expansion during the 2000s and early 2010s gave way to a more controlled deployment path in the mid-2010s, in part because of rising congestion-management costs. Since 2021, policy has shifted sharply back toward expansion, with statutory targets requiring around 87 percent more wind capacity by 2030. The long history of deployment is useful for the empirical analysis because I observe twenty entry cohorts between 2004 and 2023, while EEG support is designed around a twenty-year remuneration horizon. I can therefore measure the expected lifetime value relevant for entry rather than relying on contemporaneous annual revenues.

The German transmission system also illustrates why renewable siting and network investment have to be considered jointly. Much of the existing high-voltage network was built around a generation system dominated by coal and nuclear plants, whereas the energy transition has shifted an increasing share of generation toward wind-rich northern and eastern regions. The major north–south transmission projects evaluated later in the paper were already being planned in the early 2010s in response to this changing geography of generation. Their long planning and construction periods highlight an important asymmetry between the two policy margins studied here: wind investment can respond to current incentives well before the transmission network can adjust.

Against this background, I focus on the spatial incentives embedded in the Renewable Energy Sources Act (EEG), in particular the reference yield model, which adjusts support according to local wind conditions. I therefore turn next to the evolution of wind remuneration under the EEG.

3.1 Wind remuneration under the EEG

Wind investment in Germany has been supported through the EEG throughout the period I study. The precise rules changed repeatedly, from administratively set feed-in tariffs to direct marketing and, eventually, auctions. For the analysis, these institutional changes serve one main purpose: they map a project's entry cohort, location and year together with the wind turbine model into its expected lifetime revenue. I implement the applicable rules cohort by cohort and provide the full remuneration schedules in the Appendix 7.2.

To fix notation, let g denote a location and c the year in which a project j enters, which I refer to as its entry cohort. Let $t \in \{1, \dots, 20\}$ denote a calendar year during the project's operating life. The twenty-year horizon corresponds to the common support duration under EEG. I write q_{jcgt} for expected electricity production in remuneration year t which can be decomposed into the hourly raw wind speed vector $\mathbf{w}_{gt}$ mapped by the power curve $F_j(.)$ of the wind turbine model. We have s_{jcgt} for total remuneration and hence annual revenue is

$$r_{jcgt} = q_{jcgt} s_{jcgt} = F_j(\mathbf{w}_{gt}) s_{jcgt}, \tag{1}$$

and the expected discounted lifetime revenue used in the entry model is

$$V_{jcg} = \sum_t \beta^{t-1} \mathbb{E}[r_{jcgt}] = \sum_t \beta^{t-1} \mathbb{E}[q_{jcgt} s_{jcgt}]. \tag{2}$$

The remuneration term s_{jcgt} accommodates both the earlier feed-in-tariff system and later direct marketing. Under a fixed feed-in tariff, it is simply the applicable statutory payment a_{jcgt}. Under direct marketing, the operator sells its output on the electricity market and receives a market premium whenever the applicable EEG value exceeds the relevant wind market value. I write

$$s_{jcgt} = p_{jt} + \max\{a_{jcgt} - m_t, 0\}, \tag{3}$$

where m_t is the expected wind market value in remuneration year t and p_{jt} is the electricity price at which the energy is sold. The wind market value is calculated monthly by the TSOs and captures the average electricity price at which wind energy has sold. Total remuneration equals the applicable value plus a spread for profitably selling the electricity $p_{jt} - m_t$ whenever the market premium is positive. If the market revenue exceeds that value then a project simply gets the electricity price p_{jt}. The auction system introduced an auction into direct marketing. The statutory payment a_{jcgt} is consequently auctioned off.

Reference yield model. A central spatial feature of onshore support is the reference yield model, which adjusts remuneration for the quality of the wind site. The reference yield is the electricity that the same turbine type and hub height would produce over five years at a standardized statutory reference site. It therefore makes the applicable EEG value a_{jcgt} project and location

specific. Let Q_{jcg}^{site} denote expected five-year production at the actual site and Q_{jcg}^{ref} the corresponding reference yield. I denote the quality factor by

$$\phi_{jcg} = 100 \frac{Q_{jcg}^{\text{site}}}{Q_{jcg}^{\text{ref}}}. \tag{4}$$

A value below 100 therefore describes a site with lower expected production than the reference site. The physical definition of the reference site changed over time, most importantly with the introduction of the auction regime, and I use the definition applicable to each entry cohort.

Before auctions, onshore projects generally received a relatively high initial payment followed by a lower basic payment. The quality factor determined how long the higher payment remained in place: projects at weaker-wind sites retained the initial remuneration for longer. The reference yield model therefore partly compensated locations with lower wind productivity.

The auction regime introduced from 2017 changed both how the level of support is determined and how the reference-yield adjustment operates. Bids are stated for a project at the 100-percent reference site. If z_j denotes the relevant auction value for project j at that reference site, the site-specific applicable value in remuneration cohort c can be written schematically as

$$a_{jcg} = z_j \Gamma_c(\phi_{jcg}), \tag{5}$$

where $\Gamma_c(.)$ denotes the cohort-specific statutory correction schedule. The correction is greater than one at weaker sites and below one at sufficiently strong sites. The auction value is adjusted for site quality and the resulting applicable value governs the constant remuneration over the full support horizon. Thus, the reform can be viewed as a move from adjusting the *duration* of the high tariff to adjusting the *level* of a single applicable value.

For this paper, the important feature is that this adjustment is limited. The reference yield model makes low-wind locations more attractive than they would be under a geographically uniform payment, but it does not equalize expected revenue across sites. More importantly, the correction is based on wind productivity, not on the value of the resulting electricity to the transmission system. Neither ϕ_{jcg} nor $\Gamma_c(\phi_{jcg})$ directly incorporates local grid congestion, redispatch costs, or the marginal value of an additional MWh of wind at location g.

Congestion and private wind revenues. The institutional rules above matter for measuring the expected lifetime revenue V_{jcg}, but they provide no direct congestion signal to wind investors. This distinction is particularly important in Germany because congestion can make an additional MWh of wind substantially less valuable to the electricity system in one part of the grid than in another.

Other elements of market design further insulate private project revenues from local congestion. Germany operates as a single bidding zone, so internal transmission constraints do not translate into locational wholesale prices (which could enter through the direct marketing regulation), and

purchasing power agreements are uncommon over the study period. Renewable generation has priority in dispatch, and grid-induced curtailment has historically been largely compensated. Under the historical curtailment rules, plants commissioned from 2012 onward were generally compensated for 95 percent of revenues lost through grid-induced curtailment; once forgone revenues in a calendar year exceeded one percent of annual revenues, further forgone revenues were compensated at 100 percent. Older plants were compensated at 100 percent. From October 2021, renewable curtailment was incorporated into the broader Redispatch 2.0 framework, which introduced balancing and financial compensation for affected plants. The exact compensation mechanism therefore changes over the sample, but the economic point is the same: virtually all of private revenue loss created by network congestion is compensated rather than borne by the wind producer.

Grid-connection rules reinforce the same separation. Developers have a statutory right to grid connection and bear the cost of connecting a project to the legally determined connection point, whereas the costs of optimizing, reinforcing, and expanding the existing network are borne by the network operator. Wind developers therefore face a largely shallow connection-cost regime rather than the full network cost generated by their location choice.

Together, these rules imply that spatial differences in expected revenue V_{jcg} arise primarily through wind productivity and the design of EEG remuneration rather than through a direct price for local transmission congestion. The temporary grid expansion area between 2017 and 2020 is an important exception: it restricted annual auction awards in congested northern regions to 58 percent of average entry over 2013–2015, but did so through a coarse quantity restriction rather than a continuous congestion price.

The outlined regulatory separation is useful empirically because project revenues vary substantially across space while local congestion itself is largely excluded from private wind revenues.

4 Model

The model has two building blocks. First, I model how expected lifetime revenue determines the amount of onshore wind capacity entering a location. Second, I model short-run electricity dispatch under grid constraints to quantify how the resulting spatial distribution of wind capacity affects generation costs, redispatch, and curtailment.

4.1 Wind Turbine Entry

I model onshore wind deployment as the outcome of competitive entry by wind developers. The purpose of the model is to recover the intensive-margin response of local wind investment to changes in expected lifetime revenue. Locations are German counties. Consistent with the notation introduced in Section 3.1, let g index locations and c index entry cohorts, where $c \in \{2004, \ldots, 2023\}$.

Descriptive entry patterns. Table 2 summarizes entry over the sample period. Conditional on positive entry, the average onshore county–cohort cell receives 19.46 MW of new capacity and has expected discounted lifetime revenue of approximately EUR 3.17 million per MW. Entry is sparse: among the county–cohort cells represented in the completed panel, there are 3,433 onshore observations with zero entry, compared with 2,587 positive-entry observations. Offshore wind is shown in the table only for descriptive comparison and is not part of the entry model estimated below. There are a few reasons for this. First, there is very few cohorts in which there is entry making estimation challenging. Second, current policy with its reference yield model focuses on onshore wind turbines which is what I am to change in counterfactuals. Third, substitution between onshore and offshore projects is unlikely to be a large factor given that project scope and complexity are very different. This is also reflected in the average revenues.

Table 2. Summary statistics for wind turbine entry

Sample	Average entry (MW)	Average lifetime revenue (EUR/MW)	Positive entry	Zero entry
All	22.47	3,180,280	2,601	3,439
Onshore	19.46	3,168,846	2,587	3,433
Offshore	579.41	5,293,210	14	6

The large number of zero-entry observations reflects both economically unattractive locations and locations in which development may be limited by geography, land-use regulation, permitting, or other constraints. Of the 400 German counties 98 do not see any wind turbine entry. Among these 85 are urban counties according to the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR). The remaining counties are almost exclusively in Bavaria which is know for its strict rules on wind turbine entry. Rather than imposing a common model on both the extensive and intensive margins, I condition the estimation on county–cohort cells with positive observed entry. The model therefore asks how much capacity enters once a location is active, not whether a previously inactive location becomes available for wind development. I believe this assumption to be better than the alternative of using assumptions to derive revenue in zero entry counties because most of these counties are having high population density or politically unfavorable conditions. It is unlikely that a counterfactual under the same budget constraint would lead to sudden exploration of these areas.

Project value and local entry. For each turbine model j entering location g in cohort c, Section 3.1 defines expected discounted lifetime revenue V_{jcg}. I aggregate these project-level values to the county–cohort level and treat the resulting capacity-weighted average as the value of a representative MW of entry in that cell. The entry model therefore abstracts from within-cell heterogeneity and assumes homogeneous entry within a county–cohort cell. Allowing for heterogeneous technology in the model is beyond the scope of this paper.

Let $\mathcal{J}_{cg}$ denote the set of turbines entering in (c, g) and let K_j denote turbine j's capacity in MW. Aggregate entry is

$$k_{cg} = \sum_{j \in \mathcal{J}_{cg}} K_j, \tag{6}$$

and expected lifetime revenue per MW is

$$v_{cg} = \frac{\sum_{j \in \mathcal{J}_{cg}} V_{jcg}}{k_{cg}}. \tag{7}$$

Thus, v_{cg} is the expected operating value per MW assigned to the representative project in a county–cohort cell. Differences in turbine models, hub heights, local wind conditions, and the applicable remuneration rules enter through the project-level values constructed in the preceding section before being aggregated into this representative value. This creates substantial variation in lifetime revenue across both space and cohorts. Importantly, the institutional separation described in Section 3.1 means that this variation is driven primarily by wind productivity, technology, and the statutory remuneration schedule rather than by local congestion itself.

I assume competitive entry by atomistic developers with rational expectations. Within a county–cohort cell, developers are homogeneous in the entry model and earn the same expected lifetime revenue per MW v_{cg}. They take expected remuneration and prevailing local input prices as given. Development requires locally scarce inputs such as suitable land, grid-connection opportunities, permitting capacity, and other development services. As aggregate entry increases, competition for these inputs raises their equilibrium price and hence the common development cost per MW faced by developers. I summarize this relationship by the equilibrium unit development-cost schedule

$$c_{cg}(k) = \mu_g + \delta_c + \lambda k, \qquad \lambda > 0, \tag{8}$$

where μ_g captures persistent regional differences in development conditions, while δ_c absorbs nationwide changes in turbine investment costs, financing conditions, permitting, and other common determinants of entry costs. All active developers face the same equilibrium cost per MW, which rises with aggregate entry because local development inputs are scarce.

Under perfect competition and free entry an active county–cohort cell therefore satisfies

$$v_{cg} = c_{cg}(k_{cg}), \qquad k_{cg} > 0. \tag{9}$$

This is a zero-profit condition for the representative homogeneous project. Combining equations (8) and (9) yields the inverse entry equation that I bring to the data

$$v_{cg} = \mu_g + \delta_c + \lambda k_{cg} + \varepsilon_{cg}, \qquad k_{cg} > 0. \tag{10}$$

The parameter λ measures how quickly the common equilibrium development cost per MW rises with aggregate local entry. Equivalently, $1/\lambda$ is the intensive-margin response of entry to a change

in expected lifetime value per MW. I collect remaining county–cohort-specific development costs in ε_{cg}.

The identifying assumption is that, conditional on regional and cohort fixed effects, remaining county–cohort-specific development-cost shocks are orthogonal to observed entry. The German institutional setting removes an important potential confounder from this relationship. Because wind producers face a national wholesale price, are largely compensated for grid-induced curtailment, and do not pay for reinforcement of the existing transmission network, locations with better grid access do not mechanically receive higher private operating revenues. Hence the revenue variation used to estimate λ is not itself constructed from the local congestion outcome that I later introduce in the counterfactual. This does not eliminate all possible correlation between unobserved development costs and entry, but it makes the estimated intensive-margin response particularly transparent for the policy experiment: the LMP counterfactual changes project value through a new congestion component that is absent from the status-quo revenue formula.

Estimation. I estimate equation (10) on the 2,587 onshore county–cohort observations with positive entry. Table 3 reports specifications with progressively richer fixed effects. The estimated relationship between lifetime revenue and entry is positive throughout. My preferred specification is column (4), which includes both NUTS2 region and cohort fixed effects. It yields

$$\widehat{\lambda} = 1{,}208.8. \tag{11}$$

The units are EUR of lifetime revenue per MW for one additional MW of local entry. Hence, conditional on NUTS2 and cohort effects, an increase of EUR 100,000 in expected lifetime revenue per MW corresponds to approximately 83 MW of additional entry on the intensive margin.

Table 3. Onshore entry model with cohort and NUTS2 fixed effects

Dependent Variable:	Lifetime revenue per MW			
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	3,074,169.3***			
	(18,101.3)			
Entry (MW)	4,865.5***	3,069.1***	3,479.9***	1,208.8***
	(703.7)	(537.7)	(558.6)	(331.3)
Fixed effects				
NUTS2		Yes		Yes
Cohort			Yes	Yes
Fit statistics				
Observations	2,587	2,587	2,587	2,587
R^2	0.05699	0.27694	0.24044	0.48523
Within R^2		0.02621	0.03526	0.00548

Heteroskedasticity-robust standard errors in parentheses
*Significance: *** 1%, ** 5%, * 10%*

The fixed effects substantially reduce the estimated slope relative to the unconditional relationship. This is consistent with both entry and lifetime revenue being systematically higher in some regions and cohorts. The preferred estimate therefore uses variation in entry and project value after removing persistent NUTS2 regional differences and common cohort-level changes.

Counterfactual entry. The counterfactuals change expected lifetime project value through the spatial design of wind support while holding the local equilibrium development-cost schedule fixed. Let v_{cg}^s denote expected lifetime revenue per MW under subsidy regime s. Subtracting the status-quo entry condition from the counterfactual condition gives the intensive-margin response

$$k_{cg}^s = \max\left\{0,\ k_{cg} + \frac{v_{cg}^s - v_{cg}}{\widehat{\lambda}}\right\}. \tag{12}$$

This differencing is useful because the region effect, cohort effect, and county–cohort-specific component of development costs cancel as long as the policy does not itself change the underlying development-cost schedule.

As previously mentioned I apply equation (12) only to county–cohort cells with positive observed onshore entry. A policy can reduce an active cell to zero, but a county–cohort cell with no observed entry remains at zero. When constructing v_{cg}^s, I hold expected generation per MW and the observed composition of projects within each active cell fixed, so that the policy changes project value only through remuneration and then changes installed capacity through equation (12). Offshore entry is

kept fixed.

4.2 Electricity Dispatch

I next map a spatial allocation of wind capacity into electricity-market outcomes. The German electricity market is well suited to a two-stage representation. Electricity is first traded in a single national bidding zone without internal transmission constraints. After market clearing, transmission-system operators adjust generation when the resulting injections are incompatible with the physical grid. I mirror this sequence. In Stage 1, conventional generation is dispatched nationally according to the merit order. In Stage 2, generation is redispatched across the four German transmission-system-operator (TSO) areas—50Hertz, TenneT, Amprion, and TransnetBW—subject to the transmission network. The county assignment to each TSO is shown in Figure 3. Wind is curtailed only when redispatch alone cannot produce a feasible allocation. This distinction is central for the counterfactuals: changing the location of wind can leave nationwide generation almost unchanged while substantially changing the amount of redispatch and curtailment required to operate the grid.

The dispatch model evaluates each wind allocation in the same 2023 electricity-system environment. Demand, net imports, non-wind renewable generation, storage operation, the conventional generation fleet, marginal generation costs, and the transmission network are held fixed across subsidy regimes. Offshore wind is also held fixed. The only object that changes with subsidy design is the spatial stock of onshore wind implied by the entry model. I therefore interpret differences across scenarios as the short-run system consequences of reallocating onshore wind capacity, rather than as a forecast of a new general equilibrium in the electricity market.

Representative hours and electricity supply. Let $r \in \mathcal{R}$ index the four TSO areas and $h \in \mathcal{H}$ index hours. Solving the dispatch for all 8,760 hours under every policy is computationally costly, especially when electricity pricing is embedded in the LMP-adjusted subsidy counterfactual. I therefore select 100 representative 2023 hours using k-medoids clustering. The clustering uses, for each TSO, pre-wind residual demand and gross wind generation, giving eight system-state variables in total. Because medoids are actual hours rather than synthetic averages, each selected observation preserves the joint realization of demand, wind, and other system conditions. I detail the precise implementation of the clustering in Appendix 7.3. Let n_h denote the number of 2023 hours represented by medoid h, so that $\sum_h n_h = 8{,}760$.

Let D_{rh} denote observed electricity load, augmented by pumped-storage charging. Let $\widetilde{D}_{rh}$ be domestic demand as D_{rh} minus exogenous net imports. I take net imports as fixed because the object of interest is congestion within Germany rather than endogenous adjustment at international interconnectors. Let R_{rh} collect all other fixed low-marginal-cost generation, including solar, biomass, hydro, other renewables, and observed pumped-storage generation. Storage charging and generation are therefore held at their observed 2023 levels rather than optimized intertemporally.

Let $i \in \mathcal{I}_r$ index conventional generating units in TSO r. Unit i has marginal cost m_i and available dispatch capacity $\overline{G}_i$. Marginal cost combines fuel and transport costs, plant efficiency, carbon costs, and variable operating costs,

$$m_i = \frac{f_i + \tau_i}{\eta_i} + p^{CO_2} \frac{e_i}{\eta_i} + u_i. \tag{13}$$

Here, f_i is the fuel price per MWh of thermal input, τ_i is the corresponding fuel-transport cost, η_i is plant efficiency, p^{CO_2} is the EU ETS allowance price per tonne of CO_2, e_i is the fuel-specific emissions factor per MWh of thermal input, and u_i is the non-fuel variable operating cost per MWh of electricity. Hence m_i is measured in EUR/MWh of electricity. I assume competitive cost-based dispatch and abstract from strategic bidding.

To account for conventional-plant availability without imposing realized hourly outage patterns, I assign each energy source fixed average unavailability rate and apply the same rate to all plants using that source. Available capacity is therefore time invariant across representative hours and identical across subsidy scenarios. It varies across plants through net nominal capacity and across technologies through the source-specific unavailability rate. This abstracts from transitory plant-level outage shocks and keeps the conventional supply side fixed when comparing alternative spatial allocations of wind. Generation satisfies

$$0 \leq G_{ih} \leq \overline{G}_i. \tag{14}$$

Each representative hour is solved independently. I therefore abstract from start-up costs, minimum-run constraints, ramping, and intertemporal storage arbitrage. These restrictions matter for a detailed reconstruction of unit commitment, but are less central for the object here: the relative congestion consequences of changing the geographic distribution of wind while holding the rest of the electricity system fixed.

Hourly wind production. The entry model changes wind capacity at the county–cohort level, while dispatch requires hourly generation. I construct a common hourly productivity profile for each active county–cohort cell and hold it fixed across policy scenarios. Continuing the notation from the entry model, let g denote a county, c an entry cohort, and $\mathcal{J}_{cg}$ the set of observed turbines in that cell. For turbine j, I evaluate realized 2023 wind conditions at its location and scale wind to turbine hub height. A direct mapping of raw reanalysis wind speeds through turbine power curves tends to overpredict realized wind generation, a well-known issue in wind-generation simulation (Geiger et al., 2026; Gruber et al., 2019). I therefore apply a multiplicative wind-speed correction, calibrated by TSO and technology against hourly 2023 observed generation plus curtailment, before mapping wind speed through the turbine-specific power curve. The details are in Appendix 7.1.Let q_{jh} denote the resulting gross output in hour h and K_j the turbine's installed capacity. The county–cohort

capacity factor is

$$CF_{cgh} = \frac{\sum_{j \in \mathcal{J}_{cg}} q_{jh}}{\sum_{j \in \mathcal{J}_{cg}} K_j}. \tag{15}$$

The wind field used here is realized hourly 2023 ERA5 weather rather than the long-run-normalized wind distribution used to form expected lifetime revenues. For the 2023 cohort, output is set to zero in hours preceding each turbine's observed commissioning date, preserving the within-year commissioning pattern.

Let k^s_{cg} denote installed onshore capacity under subsidy regime s. Gross onshore wind in TSO r is then

$$W^{s,\text{on}}_{rh} = \sum_{g:r(g)=r} \sum_{c} CF_{cgh} k^s_{cg}, \tag{16}$$

and total gross wind is

$$W^s_{rh} = W^{s,\text{on}}_{rh} + W^{\text{off}}_{rh}, \tag{17}$$

where offshore wind is common across scenarios. Hence a subsidy counterfactual scales the amount of onshore capacity in existing county–cohort cells while holding fixed their observed site and technology composition and their hourly wind productivity. Historical curtailment is not added to W^s_{rh}; it is used only to reconstruct gross historical wind when selecting and validating the representative hours. Curtailment is determined endogenously below.

Stage 1: national market dispatch. Stage 1 represents market clearing in Germany's single bidding zone. Internal transmission constraints are ignored and wind and other fixed generation enter ahead of conventional generation. The national residual requirement for conventional generation is

$$Q^s_h = \sum_r \left(\widetilde{D}_{rh} - R_{rh} - W^s_{rh} \right). \tag{18}$$

If $Q^s_h > 0$, conventional units are dispatched in increasing order of m_i according to

$$\begin{aligned} \min_{\{G_{ih}\}} \quad & \sum_i m_i G_{ih} \\ \text{s.t.} \quad & \sum_i G_{ih} = Q^s_h, \\ & 0 \le G_{ih} \le \overline{G}_i. \end{aligned} \tag{19}$$

Let $G^{1,s}_{ih}$ be the solution of the above problem under subsidy regime s. If gross fixed generation exceeds national demand, conventional generation is set to zero and the excess is removed in the subsequent minimum-curtailment step. Let $G^{1,s}_{rh} = \sum_{i \in \mathcal{I}_r} G^{1,s}_{ih}$ denote Stage 1 conventional generation in TSO r and

$$C^{1,s}_h = \sum_i m_i G^{1,s}_{ih} \tag{20}$$

denote generation cost without internal grid constraints.

Stage 2: congestion management. Stage 2 asks how closely the market allocation can be implemented on the physical transmission network. The four TSO areas are the network nodes. I retain physical transmission lines that connect different TSOs and abstract from bottlenecks within a TSO. Let κ_{rh} denote wind curtailment. Net injection into the transmission network is

$$I^s_{rh} = G_{rh} + R_{rh} + W^s_{rh} - \kappa_{rh} - \widetilde{D}_{rh}. \tag{21}$$

I use a linear real-power representation of the grid which I detail in Appendix 7.4. If H maps TSO injections into physical cross-TSO line flows, flow on line ℓ is

$$Z^s_{\ell h} = \sum_r H_{\ell r} I^s_{rh}, \qquad -\overline{Z}^s_\ell \le Z_{\ell h} \le \overline{Z}_\ell, \tag{22}$$

together with national balance $\sum_r I^s_{rh} = 0$. I therefore treat the grid as a four-node transmission system with observed cross-TSO transfer constraints. I abstract from within-TSO congestion, reactive power, and voltage constraints.

I do not add a separate endogenous loss function to the transmission equations. The observed total-load measure already includes transmission and distribution losses, so the realized 2023 loss component is embedded in D_{rh}. Because observed demand is held fixed across subsidy scenarios, these losses are held fixed as well. The linear transmission model itself is lossless, however, so the counterfactuals abstract from changes in electrical losses that would result from changing the geographic pattern of power flows.

Renewable priority is represented by making wind curtailment a last-resort feasibility instrument. Before final redispatch, I find the smallest aggregate amount of wind that permits a feasible network allocation,

$$\min_{\{\kappa_{rh}\},\{G_{rh}\}} \sum_r \kappa_{rh}, \qquad 0 \le \kappa_{rh} \le W^s_{rh}, \tag{23}$$

subject to conventional capacity, national balance, and the line constraints in equation (22). Among allocations requiring the same minimum curtailment, I select the one requiring the least redispatch. Curtailment compensation does not enter this optimization: wind is curtailed only to restore physical feasibility, not because curtailment is monetarily cheaper than conventional redispatch.

Given the resulting κ^s_{rh}, the final Stage 2 allocation minimizes the absolute TSO-level deviation from the Stage 1 market outcome,

$$\min_{\{G_{rh}\}} \sum_r \left| G_{rh} - G^{1,s}_{rh} \right|, \tag{24}$$

subject to the same generation, balance, and transmission constraints with $\kappa_{rh} = \kappa^s_{rh}$. Conditional on the required generation in each TSO, conventional units within the TSO are again dispatched in

merit order. This objective reflects the institutional sequence in which the national market clears first and the system operator subsequently changes that market schedule only as much as necessary to operate the network. I do not introduce an artificial backstop technology. An hour that cannot be served with the observed fleet and permitted wind curtailment remains infeasible and is reported as such.

Let $G_{rh}^{2,s}$ denote the resulting unit-level dispatch. Generation cost with the grid constraint is

$$C_h^{2,s} = \sum_i m_i G_{rh}^{2,s}. \tag{25}$$

I measure one-sided redispatch volume as

$$RD_h^s = \frac{1}{2} \sum_r \left| G_{rh}^{2,s} - G_{rh}^{1,s} \right|, \tag{26}$$

and redispatch cost as the change in conventional generation cost,

$$RC_h^s = C_h^{2,s} - C_h^{1,s}. \tag{27}$$

Finally, I attach an ex-post compensation measure to curtailed wind. Let $\bar{s}_{rh}^s$ denote the average remuneration of supported wind generation in TSO r and hour h and subsidy regime s. Curtailment cost is

$$CC_h^s = \sum_r \bar{s}_{rh}^s \kappa_{rh}^s, \tag{28}$$

and total system cost reported in the results is

$$TC_h^s = C_h^{2,s} + CC_h^s. \tag{29}$$

I also report $RC_h^s + CC_h^s$ to isolate the congestion-management cost relative to the unconstrained national market outcome.

Locational marginal prices. The Stage 2 objective minimizes changes from the market schedule rather than generation cost itself, so its dual variables do not have the usual interpretation as locational marginal prices. I therefore recover prices from a separate least-cost dispatch problem evaluated on the same hourly system state. The network-constrained pricing problem minimizes $\sum_i m_i G_{ih}$ subject to

$$\widetilde{D}_{rh} - G_{rh} - R_{rh}^0 - W_{rh}^s + \kappa_{rh} + I_{rh} = 0 \qquad \forall r, \tag{30}$$

national balance $\sum_r I_{rh} = 0$, the line constraints in equation (22), and the generation and wind-curtailment bounds used above.

Let ν_{rh}^s be the multiplier on equation (30), μ_h^s the multiplier on national balance, and $\eta_{\ell h}^{+,s}, \eta_{\ell h}^{-,s} \geq 0$ the multipliers on the upper and lower transmission constraints. Omitting only the terms associated

with generation and curtailment bounds, the Lagrangian is

$$\mathcal{L}_h^s = \sum_i m_i G_{ih} + \sum_r \nu_{rh}^s \left(\widetilde{D}_{rh} - G_{rh} - R_{rh} - W_{rh}^s + \kappa_{rh} + I_{rh}^s \right) - \mu_h^s \sum_r I_{rh} + \sum_\ell \eta_{\ell h}^{+,s} \left(\sum_r H_{\ell r} I_{rh}^s - \overline{Z}_\ell \right) + \sum_\ell \eta_{\ell h}^{-,s} \left(- \sum_r H_{\ell r} I_{rh}^s - \overline{Z}_\ell \right) + \mathcal{B}_h^s, \tag{31}$$

where $\mathcal{B}_h^s$ collects the generation- and curtailment-bound terms. If $C_h^s(\widetilde{D}_h)$ denotes the minimized generation cost, the envelope theorem identifies the nodal-balance multiplier with the locational marginal price,

$$p_{rh}^s \equiv \nu_{rh}^s = \frac{\partial C_h^s(\widetilde{D}_h)}{\partial \widetilde{D}_{rh}} = \left. \frac{\partial \mathcal{L}_h^s}{\partial \widetilde{D}_{rh}} \right|_* . \tag{32}$$

Thus p_{rh}^s measures the increase in minimum feasible generation cost from one additional MWh of demand in TSO r. The first-order condition for the TSO injection gives

$$0 = \frac{\partial \mathcal{L}_h^s}{\partial I_{rh}} = \nu_{rh}^s - \mu_h^s + \sum_\ell H_{\ell r} \left(\eta_{\ell h}^{+,s} - \eta_{\ell h}^{-,s} \right), \tag{33}$$

so that

$$p_{rh}^s = \nu_{rh}^s = \mu_h^s - \sum_\ell H_{\ell r} \left(\eta_{\ell h}^{+,s} - \eta_{\ell h}^{-,s} \right) \equiv \mu_h^s + \xi_{rh}^s, \tag{34}$$

where μ_h^s is the common system component and ξ_{rh}^s is the local congestion component under the chosen reference-node normalization. When no internal line constraint binds, $\eta_{\ell h}^{+,s} = \eta_{\ell h}^{-,s} = 0$ for all ℓ and all TSO prices coincide. The individual TSO LMPs are invariant to the reference-node normalization, whereas the decomposition into μ_h^s and ξ_{rh}^s is not.

A simple measure of the strength of spatial congestion in hour h is the LMP range

$$\Delta p_h^s = \max_r p_{rh}^s - \min_r p_{rh}^s. \tag{35}$$

These locational prices later provide the congestion signal used in the LMP-based subsidy counterfactual.

Aggregation and status-quo benchmark. For any hourly outcome x_h, I report the represented-hour-weighted average

$$\overline{x} = \sum_{h \in \mathcal{H}} \omega_h x_h, \qquad \omega_h = \frac{n_h}{8{,}760}. \tag{36}$$

Annual quantities can equivalently be recovered as $\sum_h n_h x_h$. The same representative hours and weights are used for every subsidy regime, so comparisons do not reflect changes in the distribution of weather or demand.

The annual validation in Table 4 shows that hourly clustering does a good job at matching

the observed gross wind generation, curtailment as well as redispatch. The dispatch model does not match the observed 2023 levels of redispatch and wind curtailment closely but performs well matching the sum of them. This is consistent with the four-node representation resulting in the omission of within-TSO bottlenecks and the choice to model curtailment as only chosen for feasibility. I therefore use the dispatch model primarily for internally consistent comparisons across alternative wind allocations evaluated in the same electricity-system environment.

Table 4. Validation of status-quo electricity-market outcomes against observed 2023 outcomes

Outcome	Actual full 2023 (TWh)	Actual, weighted medoids (TWh)	Status-quo model (TWh)
Gross wind generation	149.53	150.00	146.65
Wind curtailment	7.23	7.02	2.68
One-sided redispatch	10.07	9.28	13.76

Notes: Actual full 2023 uses all observed hours. Actual, weighted medoids evaluates the same observed series at the 100 representative hours and weights each medoid by the number of source hours assigned to it. Status-quo model values are annualized using the same representative-hour weights. Gross wind equals observed wind injection plus observed curtailment. Observed curtailment is taken from the cleaned renewable downward-adjustment series. Observed one-sided redispatch uses Strombedingter Redispatch only and equals one half of total upward plus downward redispatch MWh.

Table 5 reports selected model-implied outcomes for the observed wind allocation before introducing the policy counterfactuals. Internal transmission constraints increase average conventional generation cost from EUR 2.06 million to EUR 2.11 million per hour and require about 1.57 GWh of one-sided redispatch. Average gross wind production is 16.74 GWh per hour, of which 0.31 GWh is curtailed. Almost all curtailment in the baseline is attributable to internal grid constraints rather than national oversupply. The corresponding locational prices are substantially lower in 50Hertz and TenneT than in Amprion and TransnetBW, with an average within-hour spread of EUR 17/MWh. These spatial price differences summarize the congestion externality that the subsidy counterfactuals seek to address.

Table 5. Status-quo electricity-market outcomes, 2023

Outcome	Status quo
Dispatch	
Generation cost w/o grid constraint (EUR/h)	2,062,390
Generation cost w/ grid constraint (EUR/h)	2,109,606
Redispatch volume, one-sided (MWh)	1,570.4
Redispatch cost (EUR/h)	47,215
Gross wind generation (MWh)	16,740.3
Wind curtailment (MWh)	305.9
of which: national balance (MWh)	5.1
of which: internal grid constraints (MWh)	300.8
Curtailment cost (EUR/h)	27,533
Total cost (EUR/h)	2,137,139
Prices	
National one-zone price (EUR/MWh)	106.01
50Hertz LMP (EUR/MWh)	96.15
TenneT LMP (EUR/MWh)	96.70
Amprion LMP (EUR/MWh)	109.64
TransnetBW LMP (EUR/MWh)	112.93
Within-hour LMP range (EUR/MWh)	17.07

Notes: Entries are represented-hour-weighted averages over 100 representative hours covering all 8,760 hours of 2023. All representative hours are feasible in the status-quo baseline. Total cost equals Stage 2 conventional generation cost plus the ex-post measure of wind-curtailment compensation. The one-zone price is obtained from the unconstrained least-cost pricing problem; TSO LMPs are obtained from the corresponding network-constrained least-cost problem.

5 Results and Counterfactuals

I use the estimated entry response and dispatch model to compare three spatial designs of onshore wind support. The first is the observed status quo described in Section 3.1, under which the reference yield model adjusts support for local wind productivity but does not price transmission congestion. The second removes this spatial adjustment and makes the supported rate geographically flat within each entry cohort thereby mimicking free entry with a guaranteed support level. The third instead adjusts the supported rate using lagged locational marginal prices, thereby exposing entry to the marginal value of electricity at different locations in the transmission system. In both counterfactuals, offshore wind is unchanged and the government's lifetime onshore subsidy budget is held fixed cohort by cohort. The exercises therefore change the spatial allocation of support rather than giving the regulator a larger subsidy budget.

5.1 Counterfactual subsidy design

Let B_c^{SQ} denote the undiscounted lifetime government expenditure on onshore wind support implied by the status-quo subsidy regime in cohort c. Let b_{cg}^s denote the corresponding undiscounted lifetime government payment per MW in county g under counterfactual subsidy regime s. Both counterfactuals satisfy

$$\sum_g k_{cg}^s b_{cg}^s = B_c^{SQ} \qquad \text{for every cohort } c. \tag{37}$$

Importantly, this is a budget constraint rather than a capacity constraint. The amount of entry can change because a different spatial allocation of the same support budget changes expected lifetime project values and hence entry through equation (12).

Geographically flat support. For the flat counterfactual, I choose one supported rate a_c^F for each entry cohort and apply it to every active onshore county. The counterfactual preserves the remuneration institution facing the project. A project under fixed remuneration receives a_c^F per MWh. A directly marketed project continues to sell electricity in the market and receives a premium only when a_c^F exceeds the expected wind market value. Thus the counterfactual removes the location-specific reference-yield adjustment but does not otherwise turn directly marketed projects into fixed-price projects.

For a candidate a_c^F, I construct the implied lifetime revenue v_{cg}^F using the same expected generation per MW and the same mix of remuneration types as in the observed county–cohort cell. Equation (12) then gives k_{cg}^F. I choose a_c^F so that the resulting government payments satisfy the governmental budget constraint in equation (37). Because the supported rate affects both subsidy payments per MW and the amount of entry receiving them, the budget condition determines the flat rate jointly with counterfactual entry.

Relative to the status quo, the flat policy removes the partial compensation that the reference yield model provides to weaker-wind sites and the partial punishment provided to stronger-wind sites. The payment rate itself no longer depends on local wind conditions, although expected project value still does because a MW installed at a windier site produces more electricity. The flat policy therefore rewards wind productivity more directly, but it remains entirely silent about whether that production is located on the valuable or congested side of the transmission network.

Sequential LMP-adjusted support. The LMP counterfactual replaces the wind-resource-based spatial adjustment with a congestion signal. Let $r(g)$ denote the TSO containing county g. I start from observed the observed stock of onshore wind turbines in 2004. Before setting support for each subsequent cohort c, I evaluate the electricity system with the counterfactual wind stock inherited from earlier cohorts and recover the represented-hour-weighted LMPs at the TSO-level

and the corresponding national one-zone price from the pricing problem in Section 4.2. I define

$$\rho_{r,c-1} = \frac{\overline{p}_{r,c-1}}{\overline{p}^0_{c-1}}, \tag{38}$$

where $\overline{p}_{r,c-1}$ is the average LMP in TSO r and $\overline{p}^0_{c-1}$ is the average one-zone price generated by the inherited wind stock. The supported rate offered to cohort c is

$$a^{LMP}_{rc} = \alpha_c \rho_{r,c-1}. \tag{39}$$

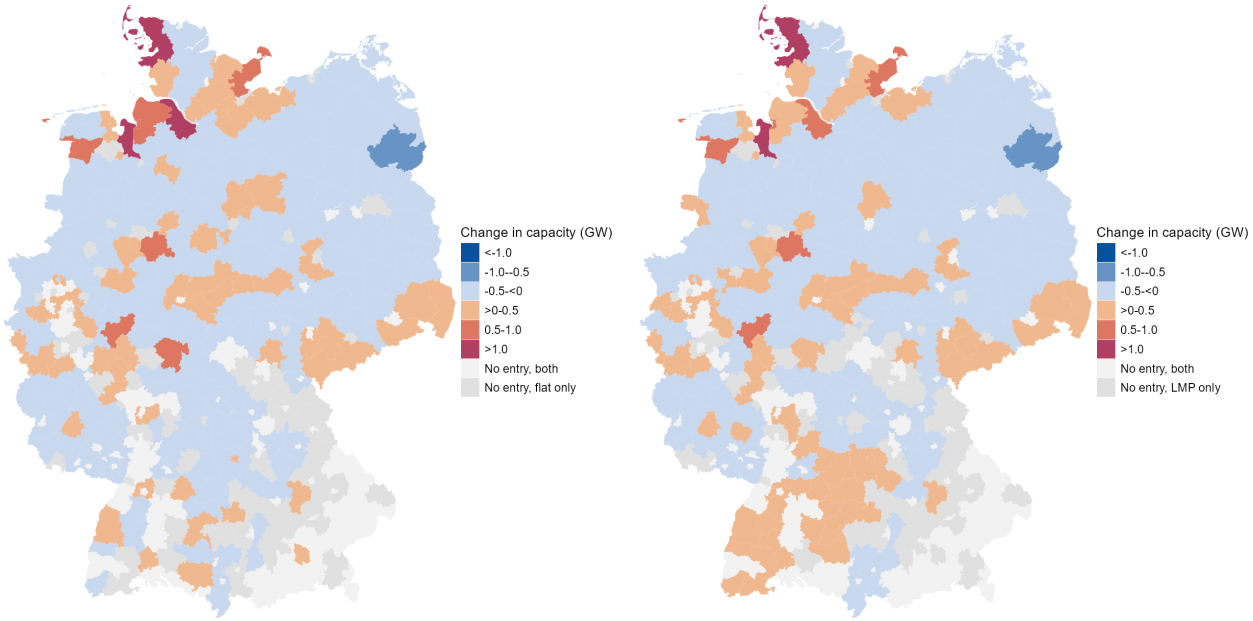
The common component α_c is chosen so that lifetime government expenditure again satisfies the budget constraint in equation (37). The same fixed-remuneration and direct-marketing rules used in the flat counterfactual then map a^{LMP}_{rc} into private lifetime revenue and government subsidy payments.

This normalization makes the policy depend on the *relative* local value of electricity rather than the overall electricity-price level. A TSO with an LMP below the national price receives a multiplier below one, while a TSO in which an additional MWh is relatively valuable receives a multiplier above one. Entry for cohort c then follows equation (12), the new capacity is added to the inherited wind stock, and the LMP signal is recomputed before the next cohort enters. The other elements of the electricity system are kept fixed at the level of early 2023 before the retirement of the last nuclear power plants. Because the dispatch model has four TSO nodes, the congestion adjustment is common to counties within a TSO; county-level entry nevertheless remains heterogeneous because wind productivity and wind turbine model mix differ across counties.

The economic contrast across the three regimes is therefore sharp. Status-quo support differentiates locations according to wind productivity, flat support does not differentiate the supported rate geographically at all, and LMP-adjusted support differentiates locations according to their marginal value to the transmission system. Both alternatives are evaluated under exactly the same cohort-specific government budget as the status quo.

5.2 Spatial reallocation of wind entry

Figure 4 shows the resulting change in total onshore capacity by county, summing entry over the 2004–2023 cohorts. The maps report counterfactual capacity minus status-quo capacity, so blue counties lose entry and orange or red counties gain entry. Grey counties with positive status-quo entry are driven to zero by the counterfactual.



(a) Flat minus status quo subsidy **(b)** LMP minus status quo subsidy

Figure 4. Change in onshore wind capacity under alternative subsidy designs

The flat policy produces substantial reallocation even though its support budget is unchanged. Removing the reference-yield adjustment reduces entry across a large number of counties while increasing it strongly in a smaller set of locations, including several high-wind coastal counties. This is the expected consequence of removing a policy that partly compensates weaker wind resources: the common payment rate interacts with differences in generation per MW and shifts investment toward locations where a MW produces more electricity. It shows that many southern regions are only sustained by the reference-yield model and consequently reduced to zero entry under flat subsidy. The pattern is not a simple north–south split, however. Wind productivity and the status-quo adjustment vary substantially within regions, so both gains and losses occur within the same TSO.

The LMP-adjusted policy adds a second force. Low constrained prices in wind-abundant parts of northern Germany reduce subsequent support there, while higher locational values make entry in southern parts of the system more attractive. Relative to the flat policy, the resulting reallocation is more clearly directed away from the low-price northern transmission corridor and toward counties in the higher-value parts of the grid. At the same time, some of the strongest wind-resource locations in the north continue to gain capacity: high physical productivity can compensate for a lower congestion multiplier. The map therefore illustrates the trade-off the LMP rule is designed to capture. Efficient siting need not mean moving all wind south; it means balancing the productivity of a site against the system value of the electricity produced there.

5.3 Planned transmission expansion

The spatial subsidy counterfactual changes where generation is built. A second policy instrument is to change the grid itself. I therefore add the major north–south HVDC projects currently planned or under construction: A-Nord, Ultranet, SuedLink Vorhaben 3, and SuedOstLink Vorhaben 5. Each project contributes 2 GW of controllable transfer capacity resulting in 8 GW of additional inter-TSO capacity in the four-node model. A-Nord connects TenneT to Amprion, Ultranet connects Amprion to TransnetBW, SuedLink Vorhaben 3 connects TenneT to TransnetBW, and SuedOstLink Vorhaben 5 connects 50Hertz to TenneT. At the level of aggregation used here, SuedLink Vorhaben 4 is not separately represented because both of its endpoints lie within TenneT.

I model these projects as additional lossless controllable transfers between their endpoint TSOs and re-solve the same 2023 dispatch problem. Their economic purpose is closely related to the spatial mismatch studied in this paper: they increase the ability to move electricity from wind-rich northern and eastern regions toward western and southern demand centers, thereby relieving internal bottlenecks, redispatch, and renewable curtailment.

The grid exercise is deliberately ex post with respect to wind entry. I first determine the wind allocation under each subsidy regime on the existing grid and then operate that same allocation with the planned grid. Hence the comparison isolates the operational value of transmission expansion conditional on where subsidy policy has caused wind to locate. It does not allow developers to anticipate the new lines and change entry in response. This distinction is useful for showing the interaction between the two policy instruments. Spatially targeted support and transmission expansion can both reduce the same congestion externality, so a subsidy that already steers entry toward high-value locations can reduce the marginal value of additional grid capacity. Conversely, a subsidy that directs more wind toward congested locations can make grid expansion substantially more valuable.

5.4 Electricity-market effects

I next operate the six combinations of subsidy regime and grid—SQ, Flat, LMP, SQ+Grid, Flat+Grid, and LMP+Grid—using the dispatch model in Section 4.2. All reported operating outcomes are evaluated on the same 100 representative hours, representing all 8,760 hours of 2023, and every representative hour is feasible in all six cases. This common sample makes differences across columns directly comparable. Table 6 reports the main dispatch outcomes.

The flat counterfactual highlights why wind productivity alone is not a sufficient criterion for siting. It produces more gross wind and lowers unconstrained national generation cost from EUR 2.062 million to EUR 2.036 million per hour. Once the grid is imposed, however, this gain is more than offset by congestion. One-sided redispatch rises from 1.57 to 2.07 GWh per hour and wind curtailment more than doubles from 306 to 635 MWh per hour. The increase is almost entirely grid-induced: internal-grid curtailment rises from 301 to 596 MWh per hour. Redispatch

Table 6. Electricity-market outcomes across subsidy regimes, with and without the planned HVDC grid, 2023

Outcome	SQ	Flat	LMP	SQ+Grid	Flat+Grid	LMP+Grid
Generation cost w/o grid constraint (EUR/h)	2,062,390	2,035,990	2,042,302	2,062,390	2,035,990	2,042,302
Generation cost w/ grid constraint (EUR/h)	2,109,606	2,121,184	2,076,306	2,076,668	2,062,167	2,052,855
Redispatch volume, one-sided (MWh)	1,570.4	2,069.6	1,345.0	537.5	874.5	409.8
Redispatch cost (EUR/h)	47,215	85,194	34,004	14,277	26,177	10,553
Wind generation (MWh)	16,740.3	17,056.4	16,960.7	16,740.3	17,056.4	16,960.7
Wind generation used (MWh)	16,434.4	16,421.5	16,722.5	16,632.6	16,837.4	16,850.6
Wind curtailment volume (MWh)	305.9	634.9	238.2	107.7	219.1	110.1
Curtailment cost (EUR/h)	27,533	58,456	22,262	9,764	19,848	10,089
Redispatch + curtailment cost (EUR/h)	74,749	143,650	56,266	24,041	46,025	20,641
Total cost (EUR/h)	2,137,139	2,179,640	2,098,568	2,086,431	2,082,015	2,062,943
Infeasible hours (percent)	0.0	0.0	0.0	0.0	0.0	0.0

Notes: Cost, redispatch, curtailment, and wind outcomes are represented-hour-weighted averages over the identical set of 100 representative hours, representing all 8,760 hours of 2023. SQ denotes the status-quo subsidy regime, Flat the geographically flat subsidy counterfactual, and LMP the sequential LMP-adjusted subsidy counterfactual. +Grid denotes operation of the corresponding wind-entry allocation with the planned HVDC grid. Total cost is constrained Stage 2 conventional generation cost plus curtailment compensation.

plus curtailment cost therefore rises from about EUR 75,000 to EUR 144,000 per hour, and total modeled electricity-system cost is around EUR 42,500 per hour above the status quo, an increase of about 1.9 percent.

The LMP-adjusted policy achieves a different combination. Gross wind production is also higher than under the status quo, so unconstrained generation cost falls, but the spatial reallocation simultaneously reduces congestion. Redispatch falls to 1.35 GWh per hour, internal-grid curtailment falls to 212 MWh per hour, and redispatch plus curtailment cost falls to about EUR 56,000 per hour. Total modeled electricity-system cost is roughly EUR 38,600 per hour below the status quo, a reduction of about 1.8 percent. Moreover, wind actually used by the system rises from 16.43 to 16.72 GWh per hour even though gross wind increases only moderately. The improvement therefore comes not only from producing wind at productive sites but also from making a larger share of that production usable by the grid. Without any grid expansion, the LMP-adjusted subsidy achieves roughly 75 percent of the cost reduction obtained by combining the status quo with the planned grid investment.

Adding the planned HVDC projects sharply reduces congestion under every subsidy regime. Under the status quo, one-sided redispatch falls from 1.57 to 0.54 GWh per hour and internal-grid curtailment from 301 to 103 MWh per hour. The reduction is even larger in absolute cost terms under the flat subsidy because that regime creates the most severe congestion on the existing network. The cost difference between flat and status-quo support largely disappears once the planned grid is added: total modeled electricity-system cost is EUR 2.082 million per hour under Flat+Grid and EUR 2.086 million under SQ+Grid. The lowest total operating cost among the six scenarios is obtained by combining LMP-adjusted support with the planned grid, at about EUR 2.063 million

per hour. The two instruments therefore remain useful together, but the incremental value of grid expansion depends strongly on the spatial incentives embedded in the subsidy.

Table 7 shows the same mechanism through prices. The national one-zone price changes across subsidy regimes because their wind allocations produce different amounts of electricity, but it is unchanged when only the internal grid is expanded because Stage 1 does not include internal transmission constraints. The TSO LMPs instead respond strongly to both policy instruments.

Table 7. Average national and locational electricity prices across subsidy regimes, with and without the planned HVDC grid, 2023

Price measure (EUR/MWh)	SQ	Flat	LMP	SQ+Grid	Flat+Grid	LMP+Grid
National one-zone price	106.01	105.03	105.06	106.01	105.03	105.06
Observed-demand-weighted average LMP	103.08	102.13	104.21	105.64	104.45	105.42
50Hertz LMP	96.15	92.73	99.54	103.85	100.78	103.96
TenneT LMP	96.70	92.15	99.13	104.08	100.79	103.97
Amprion LMP	109.64	110.64	109.35	107.48	108.55	107.06
TransnetBW LMP	112.93	119.14	110.44	107.47	108.54	106.92
Within-hour LMP range (max-min)	17.07	27.37	11.97	3.62	7.77	3.09

Notes: Entries are represented-hour-weighted averages over the same 100 representative hours as Table 6. The net-demand-weighted average LMP weights TSO prices by modeled domestic demand after subtracting net imports; the observed-demand-weighted average uses gross observed demand. The final row is the represented-hour-weighted mean of the within-hour maximum minus minimum TSO LMP.

Under the flat subsidy, the average within-hour LMP range rises from EUR 17.07/MWh to EUR 27.37/MWh. Prices fall further in 50Hertz and TenneT while TransnetBW becomes substantially more expensive, which is consistent with greater pressure on north–south transfers. The LMP-adjusted policy instead reduces the range to EUR 11.97/MWh: northern prices rise and the highest southern price falls. This is direct evidence that the entry response to the congestion signal reduces the spatial imbalance that generated the signal in the first place. The planned grid has an even larger equalizing effect. The LMP range falls to EUR 3.62/MWh under the status quo and EUR 3.09/MWh under LMP-adjusted support. Even the flat regime, which creates the greatest congestion on the existing grid, falls to EUR 7.77/MWh once the new transfer capacity is available.

Finally, Table 8 makes the interaction between subsidy design and transmission investment explicit. I define the hourly benefit of the grid as the reduction in total modeled electricity-system cost when the same wind allocation is operated with the planned HVDC links. For comparison with this flow benefit, I annualize the EUR 14.0 billion projected gross investment cost of the four modeled inter-TSO projects over 40 years at a one-percent annual discount rate. This gives an annualized cost of EUR 426.4 million, or EUR 48,673 per hour. Operating and maintenance costs are not included.

The marginal value of the grid is largest under flat support: its operating benefit is about EUR 97,600 per hour, twice the annualized capital cost under the maintained assumptions. Under the status quo the ratio is close to one, so the planned expansion is approximately cost-covering

Table 8. Hourly benefit and annualized capital cost of the planned HVDC grid across wind-subsidy regimes, 2023

Wind policy	Grid benefit (EUR/h)	Grid capital cost (EUR/h)	Benefit-cost ratio
SQ	50,708	48,673	1.04
Flat	97,625	48,673	2.01
LMP	35,625	48,673	0.73

Notes: Grid benefit is total modeled electricity-system cost without the planned HVDC grid minus total modeled electricity-system cost with the grid for the same wind-subsidy regime. Capital cost annualizes EUR 14.0 billion over 40 years at a one-percent annual discount rate and divides by 8,760 hours. Operating and maintenance costs are excluded.

under the observed wind allocation. Under LMP-adjusted support the benefit-cost ratio falls below one, so the same expansion is no longer cost-covering because part of the congestion problem has already been addressed by changing where wind enters. This should not be interpreted as evidence that congestion-adjusted subsidies eliminate the need for transmission investment. Rather, it shows that transmission expansion and spatial subsidy design are partly substitutable instruments for managing the same locational externality. The cost of evaluating either policy in isolation can therefore be misleading when the regulator controls both the incentives governing renewable siting and the capacity of the network connecting those sites to demand.

The cost measures in this section are electricity-system operating costs rather than a complete welfare measure. They include conventional generation costs and the ex-post measure of curtailment compensation, while the subsidy budget is held fixed by construction. They do not include changes in wind-development costs, consumer surplus, or other external benefits of renewable generation. The counterfactuals are therefore best interpreted as quantifying how alternative spatial subsidy rules and grid expansion change the cost of integrating a given renewable transition into the electricity system.

6 Conclusion

This paper studies the spatial allocation of wind investment and transmission capacity as a joint policy problem. In Germany, the reference yield model affects where wind enters by adjusting support to local wind conditions, while developers are largely insulated from the congestion costs created by their location choice. Transmission expansion then responds to the resulting geography of generation. I combine an estimated wind-entry model with an electricity-dispatch model under grid constraints and endogenous congestion management to evaluate these two policy margins jointly, holding the government subsidy budget fixed.

I show that subsidy design and grid expansion cannot be evaluated independently. Under the observed wind allocation, the planned grid expansion is approximately cost-covering in the model, suggesting that the current transmission expansion is closely matched to the congestion generated

under the status-quo subsidy system ex-post. This changes sharply under alternative spatial support rules. A geographically flat subsidy shifts entry toward productive wind locations but ignores whether the resulting electricity can be transported to demand. Congestion rises and the same grid expansion becomes substantially more valuable.

A congestion-adjusted subsidy has the opposite effect. By reallocating the same subsidy budget toward locations where electricity has greater marginal system value, it reduces redispatch, curtailment, and total modeled electricity-system cost by about 1.8 percent without changing the physical grid or abandoning Germany's national bidding zone. It further achieves 75 percent of the cost relief relative to the grid expansion under the status quo. It also reduces spatial price divergence substantially. Because part of the congestion problem has already been addressed through better siting, the planned grid expansion is no longer cost-covering when evaluated on the wind allocation by congestion-adjusted subsidy. Congestion-adjusted support and transmission expansion are therefore imperfect substitutes: one changes where generation is built, while the other changes how easily electricity can be moved after those investments have been made.

The results highlight the spatial allocation of renewable support as an important policy margin in its own right. The congestion-adjustment used here is deliberately coarse, distinguishing only four regions, while the existing reference yield model differentiates support at a much finer spatial level. The estimated gains from congestion-adjusted support should therefore be viewed as conservative relative to what a more granular spatial signal could potentially achieve. This matters particularly as Germany plans a large further expansion of wind capacity. Better locational incentives today can reduce the congestion created by future renewable investment and, in turn, the amount of costly transmission expansion required to integrate it.

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7 Appendix

7.1 Wind Technology and Production

The analysis uses three distinct wind-production objects. First, the statutory reference yield and site-quality factor are legal inputs to EEG remuneration. Second, the entry model requires expected physical generation over a representative operating year. Third, the dispatch model requires hourly generation in the realized 2023 weather environment. The legal object is completed from the reference-yield information described below. The two physical-generation objects instead use hourly ERA5 wind and are kept conceptually separate from the statutory reference-yield calculation.

7.1.1 Imputation of missing turbine power curves

The Marktstammdatenregister reports rated capacity, rotor diameter, and hub height but not turbine power curves. I therefore match the turbine registry to a model-specific power-curve library. Observed manufacturer–model curves are retained whenever available. Historical manufacturer labels that refer to the same turbine technology within the same corporate holding are reconciled before any imputation.

For turbine model j, let K_j denote rated capacity, D_j rotor diameter, and

$$SP_j = \frac{K_j}{\pi(D_j/2)^2} \tag{40}$$

denote specific power. If model j has no observed curve, I first search among observed curves from the same manufacturer and use another manufacturer in the same corporate holding only if no same-manufacturer donor is available. Among admissible donors, rotor diameter is the primary matching characteristic. I retain donor models d whose rotor diameter lies within two meters of the best available rotor match and then choose the donor minimizing

$$\left|\log\left(\frac{SP_d}{SP_j}\right)\right|. \tag{41}$$

The donor pool is fixed before imputation, so an imputed curve never becomes a donor for another turbine.

I convert the donor curve to a capacity-factor curve $CF_d(w)$ and rescale wind speed using the cubic relationship between swept area and wind power. The scaling factor is

$$\zeta_{jd} = \left(\frac{SP_d}{SP_j}\right)^{1/3}, \tag{42}$$

so that the target curve in the partial-load region is

$$\widehat{CF}_j(w) = CF_d(\zeta_{jd} w). \tag{43}$$

A lower-specific-power turbine therefore reaches a given capacity factor at a lower wind speed. The donor's empirical rated and cut-out behavior is retained, capacity factors are restricted to $[0, 1]$, and the final curve is normalized to peak at the target turbine's rated capacity. This procedure is close in spirit to the specific-power rescaling used by Geiger et al. (2026). It preserves observed model curves whenever possible while avoiding a single generic fleet curve; remaining differences in control strategy and environmental conditions are a limitation of the imputation (Saint-Drenan et al., 2020).

7.1.2 Legal reference yield and site-quality completion

For onshore turbines, the EEG quality factor ϕ_{jcg} defined in Equation (4) requires expected production at the actual site relative to production at the statutory reference site. Observed registry and statutory values are retained whenever available. The following procedure is used only to complete missing legal reference-yield and site-quality objects; it is not the physical generation series used for lifetime revenue or dispatch.

Let $\bar{w}^{GWA}_{jg,100}$ denote the Global Wind Atlas long-run mean wind speed at the location of turbine model j in county g. For projects governed by the pre-2017 reference-site definition, I map the 100-meter wind speed to hub height H_j with the logarithmic profile

$$\bar{w}^{GWA}_{jg,H} = \bar{w}^{GWA}_{jg,100} \frac{\log(H_j/0.1)}{\log(100/0.1)}, \tag{44}$$

while the corresponding statutory reference-site wind speed is

$$\bar{w}^{ref}_{j,H} = 5.5 \frac{\log(H_j/0.1)}{\log(30/0.1)}. \tag{45}$$

For the post-2017 auction reference site, the analogous quantities use a power-law profile with exponent 0.25 and a statutory mean of 6.45 m/s at 100 meters,

$$\bar{w}^{GWA}_{jg,H} = \bar{w}^{GWA}_{jg,100} \left(\frac{H_j}{100}\right)^{0.25}, \qquad \bar{w}^{ref}_{j,H} = 6.45 \left(\frac{H_j}{100}\right)^{0.25}. \tag{46}$$

The statutory reference-site wind speed is never adjusted to fit the data.

Because a mean wind speed alone does not determine expected turbine output, I approximate the within-location wind-speed distribution by a Rayleigh distribution when constructing this legal object. If $F_j(w)$ is the turbine power curve introduced in Section 3.1 and w is Rayleigh distributed with mean μ, expected power is

$$\bar{F}_j(\mu) = \int_0^\infty F_j(w) f_R(w; \mu)\, dw. \tag{47}$$

This is evaluated numerically on a fine wind-speed grid. The simulated five-year reference yield is

$$\widehat{Q}^{\mathrm{ref}}_{jcg} = 5 \times 8760 \times \bar{F}_j(\bar{w}^{ref}_{j,H}). \tag{48}$$

For turbines with both a reported quality factor ϕ^{obs}_{jcg} and an official turbine-specific reference yield $Q^{\mathrm{ref},obs}_{jcg}$, the implied mean site output provides a calibration target. I use a turbine for calibration only when the simulated reference yield is reasonably close to the official value,

$$0.8 \leq \frac{\widehat{Q}^{\mathrm{ref}}_{jcg}}{Q^{\mathrm{ref},obs}_{jcg}} \leq 1.2. \tag{49}$$

For each retained turbine I choose a multiplicative adjustment ϑ_{jcg} to the GWA site wind speed such that the Rayleigh-integrated power curve matches the site yield implied by $\phi^{obs}_{jcg} Q^{\mathrm{ref},obs}_{jcg}$. The search is restricted to $\vartheta_{jcg} \in [0.40, 1.25]$; boundary solutions are excluded from the spatial calibration sample.

The turbine-level corrections are noisy and multiple turbines in one wind park should not receive mechanical independent weight. I therefore winsorize the corrections at the first and ninety-ninth percentiles and collapse them to 0.01-degree spatial cells using the median. For target location g, I use the twenty nearest calibration cells with Gaussian weights

$$w_m(g) = \exp\left[-\frac{1}{2}\left(\frac{d_m(g)}{h(g)}\right)^2\right], \qquad h(g) = \max\{10, d_{20}(g)\}, \tag{50}$$

where distances are in kilometers. The weighted local estimate is shrunk toward the national median correction when nearby information is sparse or distant. Reported registry values remain unchanged; the smoothed prediction fills only missing quality factors or reference yields. The resulting correction is best interpreted as an effective site-yield correction rather than pure meteorological measurement error because it can also absorb micrositing, wake, availability, electrical, and other losses embedded in the EEG site-yield concept. Bias correction of gridded wind products using local information is common in wind-power simulation (Gruber et al., 2019).

7.1.3 Hourly ERA5 correction and physical generation

Physical generation for revenue and dispatch is constructed from hourly ERA5 wind. Raw reanalysis wind passed through turbine-specific power curves systematically differs from observed aggregate generation, so I estimate a multiplicative wind-speed correction separately for onshore and offshore wind. For onshore wind I estimate one correction factor for each of the four TSO regions. Offshore wind is grouped into the North Sea fleet connected to TenneT and the Baltic Sea fleet connected to 50Hertz, yielding two offshore correction factors. Let w_{jh} denote ERA5 wind at turbine j mapped to its hub height and let $W^{target}_{\tau rh}$ denote observed gross wind generation for technology group τ and

region r. For candidate factor γ, simulated aggregate generation is

$$W^{sim}_{\tau rh}(\gamma) = \sum_{j \in \mathcal{J}_{\tau r}} F_j(\gamma w_{jh}), \tag{51}$$

and I choose

$$\widehat{\gamma}_{\tau r} = \arg \min_{\gamma \in [0.5, 1.5]} \frac{1}{H} \sum_h \left[W^{sim}_{\tau rh}(\gamma) - W^{target}_{\tau rh} \right]^2. \tag{52}$$

The numerical tolerance is 10^{-4} and the fit is to hourly 2023 generation rather than annual totals.

The observed target is reconstructed gross generation. I start from hourly TSO-level observed onshore and offshore wind generation and add back renewable curtailment in the same TSO and hour. Because the curtailment data identify renewable curtailment but do not separately report its onshore and offshore components, I allocate curtailment using fixed technology shares. In 50Hertz and TenneT, 41 percent of renewable curtailment is assigned to onshore wind and 59 percent to offshore wind. In Amprion and TransnetBW, where no offshore fleet enters the calibration, 100 percent is assigned to onshore wind. Thus

$$W^{target,on}_{rh} = W^{obs,on}_{rh} + s^{on}_r \kappa^{hist}_{rh}, \qquad W^{target,off}_{rh} = W^{obs,off}_{rh} + s^{off}_r \kappa^{hist}_{rh}, \tag{53}$$

with $(s^{on}_r, s^{off}_r) = (0.41, 0.59)$ for 50Hertz and TenneT and $(1, 0)$ for Amprion and TransnetBW. The resulting factors therefore reconcile simulated ERA5 production with estimated gross production before curtailment. They should be interpreted as effective production corrections and are distinct from the EEG-based site-quality completion above.

Expected annual generation for the entry model is intended to reflect long-run wind conditions rather than an unusually windy or calm realization of 2023. I preserve the hourly shape of the 2023 ERA5 series but normalize its local mean to the 2008–2017 ERA5 climatology. If $\bar{w}^{2023}_j$ and $\bar{w}^{clim}_j$ denote the corresponding local means, the normalized series is

$$w^E_{jh} = w^{2023}_{jh} \frac{\bar{w}^{clim}_j}{\bar{w}^{2023}_j}. \tag{54}$$

After mapping this series to hub height and applying $\widehat{\gamma}_{\tau r}$, expected annual generation for a project j in cohort c and location g is

$$q_{jcgt} = 8760 \frac{1}{H} \sum_h F_j\left(\widehat{\gamma}_{\tau(j) r(g)} w^{E,H}_{jh} \right), \qquad t = 1, \ldots, 20. \tag{55}$$

The same expected annual output is used in every year of the twenty-year revenue horizon, abstracting from turbine degradation and secular future changes in wind conditions.

For electricity dispatch, I instead retain the realized hourly 2023 ERA5 field. I map it to hub height, apply the same technology–TSO correction, and evaluate the turbine power curve without the climatological normalization in Equation (54). Thus the revenue calculation uses a long-run-

normalized production profile whereas dispatch holds the realized 2023 weather environment fixed. For turbines commissioned during 2023, output is set to zero before the observed commissioning date.

Table 9. Calibration and validation of wind-speed corrections

Calibration group	Correction factor	Target generation (TWh)	Sim./target before	Sim./target after	Correlation before	Correlation after
Onshore – 50Hertz	0.968	38.87	1.044	0.973	0.979	0.979
Onshore – Amprion	0.886	26.43	1.241	0.959	0.977	0.972
Onshore – TenneT	0.934	52.45	1.129	0.981	0.977	0.975
Onshore – TransnetBW	0.974	3.97	1.028	0.972	0.953	0.950
Offshore – Baltic Sea (50Hertz)	0.913	5.07	1.127	0.990	0.886	0.896
Offshore – North Sea (TenneT)	0.831	22.71	1.390	1.067	0.954	0.928

Notes: The correction factor multiplies ERA5 wind speed before applying the turbine power curve. Target generation is observed 2023 wind injection plus allocated observed curtailment. The onshore correction is calibrated separately for each TSO; offshore correction is calibrated separately for the Baltic Sea/50Hertz and North Sea/TenneT groups. The simulated-to-target ratios compare annual simulated generation with the gross observed target before and after correction. Correlations compare the corresponding hourly generation profiles.

The validation Table 9 reports the six technology–region correction factors together with their hourly fit. The correction factors are below one and successful at reducing the difference to the target generation value overall but at the cost of lower hourly correlation.

7.2 Wind Remuneration and Lifetime Revenue

7.2.1 Onshore remuneration

The onshore revenue construction follows the statutory regime applicable at entry. Before auctions, the commissioning date determines the relevant EEG schedule. Once support is awarded by auction, the auction date also determines the applicable legal regime. The main implementation rules are summarized here; project-specific statutory values are assigned at the finest date frequency required by the law.

For the pre-auction reference-yield model, let ϕ_{jcg} denote the quality factor in percentage points, as in Equation (4). Under the regimes through July 2014, the high initial payment applies for

$$M_{jcg}^{old} = \min\left\{240, 60 + 2\frac{\max\{150 - \phi_{jcg}, 0\}}{0.75}\right\} \tag{56}$$

months. Under the EEG 2014, the corresponding duration is

$$M_{jcg}^{2014} = \min\left\{240, 60 + \frac{\max\{130 - \phi_{jcg}, 0\}}{0.36} + \frac{\max\{100 - \phi_{jcg}, 0\}}{0.48}\right\}. \tag{57}$$

If a tariff transition occurs within a project year, the annual applicable value is the month-weighted average of the two rates. Table 10 reports the ordinary rates used for the 2004–2016 cohorts.

Projects through July 2014 are treated as receiving the applicable fixed feed-in remuneration. From August 2014, direct marketing is used for projects above the statutory feed-in-tariff threshold:

Table 10. Ordinary onshore wind rates before the auction regime

Commissioning period	Regime	Initial value (ct/kWh)	Basic value (ct/kWh)
Jan.–Jul. 2004	earlier 2004 schedule	8.80	5.90
Aug.–Dec. 2004	EEG 2004	8.70	5.50
2005	EEG 2004	8.53	5.39
2006	EEG 2004	8.36	5.28
2007	EEG 2004	8.19	5.17
2008	EEG 2004	8.03	5.07
2009	EEG 2009	9.20	5.02
2010	EEG 2009	9.11	4.97
2011	EEG 2009	9.02	4.92
2012	EEG 2012	8.93	4.87
2013	EEG 2012	8.80	4.80
Jan.–Jul. 2014	EEG 2012	8.66	4.72
Aug.–Dec. 2014	EEG 2014	8.90	4.95
2015	EEG 2014	8.90	4.95
Jan.–Mar. 2016	EEG 2014	8.79	4.89
Apr.–Jun. 2016	EEG 2014	8.69	4.83
Jul.–Sep. 2016	EEG 2014	8.58	4.77
Oct.–Dec. 2016	EEG 2014	8.48	4.72

500 kW during 2014–2015 and 100 kW from 2016. The turbine-level capacity is used for this classification because the data do not fully identify the statutory aggregation of multiple installations.

Projects commissioned in 2017–2018 without an auction identifier and with a pre-auction remuneration assignment are treated as transition projects. They retain Equation (57) and the pre-2017 reference-site definition. The date-specific statutory values used for these projects are shown in Table 11.

Table 11. Statutory values for non-auction EEG 2017 transition projects

Commissioning period	Initial value (ct/kWh)	Basic value (ct/kWh)
Jan.–Feb. 2017	8.38	4.66
Mar. 2017	8.29	4.61
Apr. 2017	8.20	4.56
May 2017	8.12	4.51
Jun. 2017	8.03	4.47
Jul. 2017	7.95	4.42
Aug.–Sep. 2017	7.87	4.37
Oct.–Dec. 2017	7.68	4.27
Jan.–Mar. 2018	7.49	4.17
Apr.–Jun. 2018	7.31	4.07
Jul.–Sep. 2018	7.14	3.97
Oct.–Dec. 2018	6.97	3.87

For auctioned projects, bids are stated for a 100-percent reference site. Let z_j denote the value at that reference site, as in Section 3.1. When the individual award value is unavailable, I use the capacity-weighted average successful award value of the identified auction round. The site-specific applicable value is

$$a_{jcg} = z_j \Gamma_c(\phi_{jcg}), \tag{58}$$

where the correction schedule Γ_c depends on the statutory regime. Table 12 summarizes the support points. Values between support points are linearly interpolated.

Table 12. Reference-yield correction factors under the auction regimes

Regime / quality factor (%)	50	60	70	80	90	100	110	120	130	140	150
EEG 2017	–	–	1.29	1.16	1.07	1.00	0.94	0.89	0.85	0.81	0.79
EEG 2021	–	1.35	1.29	1.16	1.07	1.00	0.94	0.89	0.85	0.81	0.79
EEG 2023	1.55	1.42	1.29	1.16	1.07	1.00	0.94	0.89	0.85	0.81	0.79

Under the EEG 2017, quality factors below 70 percent use the 70-percent factor; under the EEG 2021, values below 60 percent use the 60-percent factor. Under the EEG 2023, the 50–60

percent portion of the schedule is available only to projects in the "south region"; outside that region values below 60 percent use the 60-percent factor. The EEG 2021 south region rule instead operated through auction allocation and therefore does not change the applicable value conditional on an observed award. The south region includes all of Baden-Württemberg and Saarland as well as virtually all of Bayern and the southern parts of Rheinland-Pfalz and Hessen.

A separate regional intervention applied during the first years of the auction regime through the *Netzausbaugebiet*. From 2017 through 2020, auction awards for onshore wind in a contiguous congested northern area—Schleswig-Holstein, Mecklenburg-Vorpommern, Hamburg, Bremen, and parts of northern Niedersachsen—were subject to an annual quantity cap. The objective was to slow additional wind entry in areas in which existing transmission constraints already caused substantial curtailment and redispatch while north–south grid reinforcement was still under construction. The EEG 2017 set the maximum annual award volume in the area at 58 percent of its average annual gross additions during 2013–2015. Given average additions of about 1,555 MW in those years, the resulting cap was 902 MW per year. The rule restricted where auctioned capacity could enter but did not alter the applicable remuneration of a project conditional on receiving an award; it was abolished with the EEG 2021.

The applicable auction regime is assigned by the auction date rather than the commissioning year alone. In particular, pre-2021 awards remain under the EEG 2017 schedule even if commissioned later; awards from 2021–2022 use the EEG 2021 schedule; and the EEG 2023 schedule applies to 2023 auctions. For the entry calculation, the quality factor is held at its entry-time prediction at the statutory reviews at the beginning of years 6, 11, and 16. This avoids introducing ex-post generation information to ex-ante revenue calculation used in the wind entry decision.

7.2.2 Market-value expectations and lifetime revenue

A directly marketed project sells electricity in the market and receives the EEG premium. The relevant wind market value is a generation-weighted average of wholesale prices during wind-production hours, not the unweighted spot price. For cohort c, I form the entry-time expectation from up to the five preceding annual onshore-wind market values,

$$\bar{m}_c = \frac{1}{|\mathcal{Y}_c|} \sum_{y \in \mathcal{Y}_c} m_y^{obs}, \qquad \mathcal{Y}_c = \{c-1, \ldots, c-5\} \cap \mathcal{Y}^{obs}. \tag{59}$$

When fewer than five preceding observations are available, all available preceding years are used. For projects in cohort c, I set the expected wind market value m_t introduced in Equation (3) equal to $\bar{m}_c$ in every remuneration year t. The resulting expectation is therefore held fixed over the twenty-year project horizon.

For the expected revenue calculation, the expected sales price p_{jt} is also represented by the same

wind market value $\bar{m}_c$. Hence expected total remuneration under direct marketing is

$$s_{jcgt} = \bar{m}_c + \max\{a_{jcgt} - \bar{m}_c, 0\}. \tag{60}$$

Whenever the premium is positive, total remuneration equals the applicable value. If the market value exceeds it, the premium is zero and the project retains market revenue. Fixed-tariff projects instead receive the applicable feed-in tariff directly.

Using expected annual production q_{jcgt} from Equation (55), the private lifetime operating value used in the entry model is

$$V_{jcg} = \sum_{t=1}^{20} \beta^{t-1} q_{jcgt} \frac{s_{jcgt}}{100}, \qquad \beta = 0.95. \tag{61}$$

The twenty-year horizon is common across cohorts. It standardizes the investment-value object and is distinct from the government-expenditure accounting used in the counterfactual budget below.

7.2.3 Offshore remuneration

Offshore wind is not estimated in the entry model and is held fixed in all subsidy counterfactuals, but I construct its revenue consistently for the descriptive data. The historical offshore regimes do not use the onshore quality-factor correction. Instead, older statutory schemes extend the high-payment period for projects farther from shore and in deeper water. In the regimes for which the extension applies, months of additional high remuneration are constructed from distance d_j in nautical miles and water depth h_j in meters as

$$E_j = 0.5 \lfloor \max(d_j - 12, 0) \rfloor + 1.7 \lfloor \max(h_j - 20, 0) \rfloor, \tag{62}$$

subject to the statutory regime-specific eligibility conditions.

The main legacy schedules are summarized in Table 13. The EEG 2012 and EEG 2014 compressed options front-load support; later projects move to project-specific auction values under the WindSeeG/EEG market-premium system. For auctioned projects, a project-specific award is used when observed and otherwise the identified auction round is represented by its capacity-weighted average successful award value.

Expected offshore generation uses the hourly ERA5 procedure described above rather than the old mean-wind/Rayleigh approximation. Directly marketed offshore projects use the offshore-specific market-value series, with the same trailing five-year expectation rule as in Equation (59). Offshore capacity and its hourly production profile are held fixed when comparing the onshore subsidy regimes.

Table 13. Summary of offshore remuneration regimes used in the revenue construction

Entry period	Main payment values (ct/kWh)	Treatment
2009–2011	15.00 initial; 3.50 base	EEG 2009; high-payment duration may be extended by distance/depth.
2012–Jul. 2014	19.00 compressed; 15.00 extension; 3.50 base	EEG 2012 compressed schedule when applicable.
2014–2017	19.40 compressed; 15.40 standard/extension; 3.90 base	EEG 2014, subject to transition rules for late-2014 projects.
2018–2019	18.40 compressed; 14.90 standard/extension; 3.90 base	EEG 2014 degression.
2020	13.90 standard; 3.90 base	Legacy statutory regime where applicable.
2021–2023	project auction value	WindSeeG/EEG market premium; no onshore-style quality-factor multiplier.

7.3 Representative Hours

The dispatch model uses 100 representative hours selected from the 8,760 hours of 2023. I use k-medoids so that each representative observation is an actual hour rather than a synthetic average. This preserves coherent joint realizations of demand, wind, imports, and other system conditions.

For each TSO r, modeled domestic demand is $\widetilde{D}_{rh}$ as defined in the main text: observed load, including pumped-storage charging, net of imports. Let R_{rh} denote the same fixed non-wind renewable generation used in the dispatch model. The two clustering variables for each TSO are pre-wind residual demand

$$X^D_{rh} = \widetilde{D}_{rh} - R_{rh} \tag{63}$$

and observed gross wind

$$X^W_{rh} = W^{obs,on}_{rh} + W^{obs,off}_{rh} + \kappa^{hist}_{rh}, \tag{64}$$

where historical curtailment κ^{hist}_{rh} is added back only to reconstruct the underlying wind state. It is not imposed on the subsequent dispatch problem.

With four TSOs, the hourly clustering vector contains eight variables,

$$X_h = (X^D_{1h}, \ldots, X^D_{4h}, X^W_{1h}, \ldots, X^W_{4h})'. \tag{65}$$

Each component is standardized over the full year to zero mean and unit variance. I then apply partitioning around medoids (PAM) with Euclidean distance and 100 medoids. If representative hour h represents n_h full-year hours, its weight is

$$\omega_h = \frac{n_h}{8760}, \qquad \sum_{h \in \mathcal{H}} \omega_h = 1, \tag{66}$$

which is the same weighting used in Equation (36). All subsidy and grid scenarios use the identical medoids and weights. Annual quantities are recovered by multiplying each medoid outcome by its represented-hour count n_h and summing across medoids.

Table 14. Distributional fit of representative-hour clustering targets

Feature	Full mean (GW)	Rep. mean (GW)	Full SD (GW)	Rep. SD (GW)	Full p5 (GW)	Rep. p5 (GW)	Full p50 (GW)	Rep. p50 (GW)	Full p95 (GW)	Rep. p95 (GW)
Pre-wind residual load: 50Hertz	8.50	8.57	3.28	3.08	2.84	3.38	8.51	8.41	13.72	13.14
Pre-wind residual load: TenneT	9.16	9.05	4.74	4.54	1.77	1.94	8.72	8.86	17.30	16.20
Pre-wind residual load: Amprion	16.26	16.15	4.97	4.67	8.58	8.83	16.20	15.70	24.41	23.83
Pre-wind residual load: TransnetBW	4.78	4.80	2.25	2.14	1.39	1.54	4.54	4.58	8.69	8.41
Gross wind state: 50Hertz	5.02	5.04	4.22	4.10	0.38	0.83	3.74	3.68	13.71	13.65
Gross wind state: TenneT	8.58	8.64	6.44	6.23	0.87	1.70	7.04	7.57	20.56	20.99
Gross wind state: Amprion	3.02	3.00	2.50	2.43	0.18	0.33	2.32	2.19	8.08	8.20
Gross wind state: TransnetBW	0.45	0.44	0.41	0.40	0.01	0.02	0.33	0.28	1.30	1.28

Notes: Full refers to all 8,760 hours of 2023. Rep. reports the mean, standard deviation, 5th percentile, median, and 95th percentile of the 100 medoids, weighted by the number of source hours assigned to each medoid. Pre-wind residual load equals gross observed load including storage charging minus net imports and non-wind renewable generation. Gross wind state equals observed onshore and offshore wind injection plus observed curtailment.

Table 14 shows that the represented-hour weighting preserves the central tendencies of the eight clustering variables closely and also captures their broad dispersion and tails. As modeling

congestion requires to capture the tails well the good fit in this regard is crucial. The remaining differences in some quantiles are the cost of reducing 8,760 realized hours to 100 actual medoids rather than synthetic average states.

7.4 Transmission Network and Line Capacity

The network exposition follows the spatial electricity-network formulation in Bohn et al. (1984) and the incidence-matrix representation used by Arkolakis and Walsh (2023). For the full proof of the derivation of the linear approximation to the full alternating current grid model please refer to those two references. The dispatch network has the four TSO regions as nodes, but physical lines crossing TSO boundaries are retained separately rather than collapsed to one bilateral transfer limit. Parallel lines therefore remain distinct constraints. Let A be the line–node incidence matrix after omitting a reference node and let Ω be the diagonal matrix of line admittances. The flow mapping used in the main text is

$$H = \Omega A (A' \Omega A)^{-1}, \qquad Z_h^s = H I_{-r_0,h}^s, \tag{67}$$

where $I_{-r_0,h}^s$ contains the TSO injections I_{rh}^s from Equation (21) after omitting the reference node r_0. For line ℓ,

$$\Omega_\ell = \frac{1}{\sqrt{R_\ell^2 + X_\ell^2}}. \tag{68}$$

Line limits follow a St. Clair-type loadability rule. When the observed reactance and susceptance imply a plausible surge impedance, I use

$$\mathcal{Z}_\ell^{surge} = \sqrt{\frac{X_\ell}{B_\ell 10^{-6}}}, \tag{69}$$

provided the resulting value lies between 150 and 600 ohms. Otherwise, the surge-impedance benchmark is 250 ohms for lines at or above 300 kV, 300 ohms for 200–299 kV, 350 ohms for 100–199 kV, and 400 ohms below 100 kV. Surge-impedance loading is

$$SIL_\ell = \frac{V_\ell^2}{\mathcal{Z}_\ell^{surge}}, \tag{70}$$

and the line limit is

$$\overline{Z}_\ell^{SC} = m(L_\ell) SIL_\ell, \tag{71}$$

where the St. Clair multiplier $m(L_\ell)$ declines with line length, from roughly 3 for short lines toward 0.6 at 600 miles.

Very short line segments require regularization because near-zero electrical lengths can generate extreme admittances and artificial bottlenecks after aggregating the grid to four TSO nodes. I impose a minimum electrical length of 5 km. For shorter lines, or for lines with missing or non-positive

electrical parameters, resistance and reactance are bounded using median per-kilometer values for valid lines in the same voltage class. Susceptance is made consistent with the voltage-class surge-impedance benchmark when the raw X/B combination is implausible. If a line shorter than 5 km has an observed current rating, its capacity instead uses the three-phase thermal limit

$$\overline{Z}_{\ell}^{I} = \frac{\sqrt{3} V_{\ell} I_{\ell}}{1000}. \tag{72}$$

This prevents short substation connectors from dominating the reduced network mechanically while preserving separate physical line constraints.